

High-Mass Dilepton Resonances

Search for grand unified Z' with the ATLAS detector

Bachelorarbeit aus der Physik

Vorgelegt von

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a place of mind
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Originality Statement

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Sebastian Murk

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Date

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Preliminary Remarks

Acronyms

An alphabetically sorted list of acronyms that are used throughout the thesis is provided on pages xi-xii. Bold print is used to indicate an acronym's formation if it is not deemed obvious.

Bibliography Entries

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Figures and Tables

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List of Acronyms

ALICE A Large Ion Collider Experiment.

ATLAS A Toroidal LHC Apparatus.

BAT Bayesian Analysis Toolkit.

BSM Beyond the Standard Model.

CERN European Organisation for Nuclear Research
(originally “**C**onseil **E**uropéen pour la **R**echerche **N**ucléaire”).

CL Confidence Level.

CMS Compact Muon Solenoid.

CSC Cathode Strip Chamber.

DIS Deep Inelastic Scattering.

ECal Electromagnetic Calorimeter.

EMEC Electromagnetic End-Cap.

FCal Forward Calorimeter.

GUT Grand Unified Theory.

HCal Hadronic Calorimeter.

HEC Hadronic End-Cap.

IBL Insertable B-Layer.

ID Inner Detector.

LAr Liquid Argon.

List of Acronyms

LHC Large Hadron Collider.
LHCb Large Hadron Collider beauty.
LINAC 2 Linear Accelerator 2.
LO Leading Order.
MC Monte Carlo.
MDT Monitored Drift Tube.
MS Muon Spectrometer.
NLO Next-to-Leading Order.
NNLO Next-to-Next-to-Leading Order.
PDF Parton Distribution Function.
PS Proton Synchrotron.
PSB Proton Synchrotron Booster.
QCD Quantum Chromodynamics.
QED Quantum Electrodynamics.
QFT Quantum Field Theory.
RF Radiofrequency.
RPC Resistive Plate Chamber.
SCT Semiconductor Tracker.
SM Standard Model.
SPS Super Proton Synchrotron.
SSB Spontaneous Symmetry Breaking.
SSM Sequential Standard Model.
TGC Thin Gap Chamber.
TR Transition Radiation.
TRT Transition Radiation Tracker.
VEV Vacuum Expectation Value.

Chapter 1

Introduction

The Standard Model of Particle Physics is the most successful theory in describing non-gravitational interactions to date. With the very recent discovery of a Higgs-like boson by the LHC based experiments ATLAS and CMS [1, 2], widely believed to be compatible with the Standard Model Higgs boson due to its scalar nature, it seems that the last particle predicted by the Standard Model has been found. This discovery marks a milestone in particle physics, since it is the only experimental verification of symmetry breaking, in particular the spontaneous symmetry breaking of the Standard Model gauge group, thus confirming what had been theorised a long time ago but never proven up until now.

Yet, even after the discovery, the search for physics beyond the Standard Model remains crucial, as many of its shortcomings remain unresolved. Most importantly, this includes the hierarchy problem and the astronomical evidence for dark matter.

This thesis is organised as follows: Chapter 2 gives an outline of the Standard Model of particle physics, including two Beyond the Standard Model theories which are considered in the limit setting performed in Chapter 4. Chapter 3 introduces the ATLAS experiment at the CERN Large Hadron Collider, currently the highest energy particle collider in the world. The procedures used to extract exclusion limits for the Z' mass are outlined in Chapter 4, preceded by an overview of event selection techniques and backgrounds relevant to this analysis. In particular, this thesis presents an analysis of proton-proton (pp) collisions at a centre-of-mass energy of $\sqrt{s} = 8$ TeV in the dimuon channel, corresponding to an integrated luminosity of 20 fb^{-1} recorded in Run Ib ($\sqrt{s} = 8$ TeV) of the Large Hadron Collider. In anticipation of Run II, which is scheduled to commence data taking in June 2015, proton-proton (pp) collisions at a centre-of-mass energy of $\sqrt{s} = 13$ TeV are simulated to obtain expected limits in the dielectron channel. Chapter 5 concludes with a summary of the results obtained in this analysis.

Chapter 2

Theoretical Framework

The purpose of this chapter is to provide some of the theoretical background needed for the analysis presented in this thesis. Section one is a brief outline of the Standard Model of Particle Physics. Section two discusses Beyond the Standard Model Theories, with an emphasis on models that predict high-mass dilepton resonances, i.e. particle decay signatures that the channels considered in Section 4.6 could potentially be sensitive to. Lastly, the concept of Feynman diagrams is introduced.

2.1 The Standard Model of Particle Physics

The culmination of theoretical particle physics is called the Standard Model of Particle Physics [3, 4, 5], or simply the Standard Model (SM). It strives to describe nature in terms of fundamental particles and their interactions. Fundamental (or elementary) particles are not made up of smaller constituents; they are truly fundamental (to the best of our current knowledge). An outline of the SM particles and their classifications are presented in Figure 2.1.

A first distinction is made between particles that make up matter - these are quarks and leptons¹ - and force carrier particles (gauge bosons). Quarks and leptons come in six flavours that can be arranged in three generations of doublets reflecting their intrinsic properties:

$$\begin{array}{lll} \textit{Quark Doublets:} & \begin{bmatrix} u \\ d \end{bmatrix} & \begin{bmatrix} c \\ s \end{bmatrix} & \begin{bmatrix} t \\ b \end{bmatrix} \\ \textit{Lepton Doublets:} & \begin{bmatrix} e \\ \nu_e \end{bmatrix} & \begin{bmatrix} \mu \\ \nu_\mu \end{bmatrix} & \begin{bmatrix} \tau \\ \nu_\tau \end{bmatrix} \end{array}$$

It should be noted that only the first generation (shaded grey in Figure 2.1) is needed to make up ordinary matter. The second and third generations can be thought of as

¹Due to their spin- $\frac{1}{2}$ nature, quarks and leptons may be referred to collectively as fermions.

2.1 The Standard Model of Particle Physics

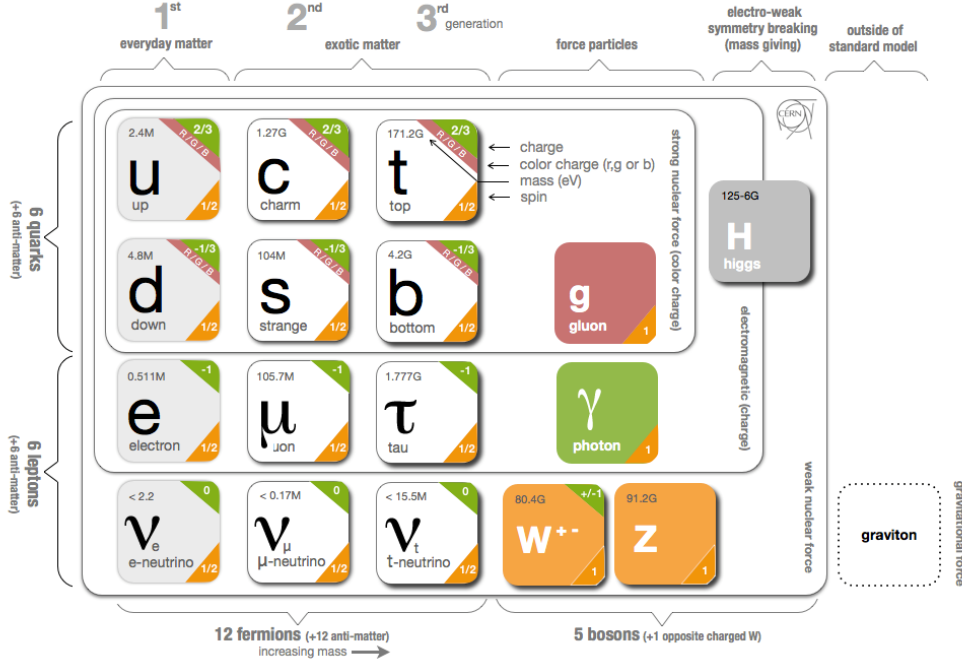


Figure 2.1: The Standard Model of elementary particles [6].

heavier, less stable duplicates of the first. They emerge from high energy collisions such as those produced in particle colliders and cosmic rays.

The mathematical formalism of the SM is a renormalisable, relativistic Quantum Field Theory (QFT) with local gauge invariance based on the gauge group

$$G_{\text{SM}} = SU(3)_C \otimes SU(2)_L \otimes U(1)_{Y_W} \quad (2.1)$$

where $SU(n)$ denotes a special unitary group of order n . Similarly, $U(n)$ represents a unitary group of order n . It incorporates three of the four fundamental forces of nature: electromagnetism, the weak interaction, and the strong interaction. Each of them is associated with one or multiple force mediating spin-1 gauge bosons. More precisely, the number of force mediating bosons associated with an interaction is given by the dimension of its conjugated Lie group. For a special unitary group $SU(n)$ of order n , the dimension is given by $n^2 - 1$. For a unitary group $U(n)$, it is simply n^2 [3, 7]. For instance, the colour symmetry group $SU(3)_C$ is of order three and represents the strong interaction. It is mediated by eight distinct types of gluons (referred to as “colour octet”). The conserved physical quantity is the colour charge denoted by the letter C . It can take on six discrete values commonly denoted red, antired, green, antigreen and blue, antiblue. Anticolours are carried only by antiparticles and vice versa. Only fermions that carry the colour charge - these are quarks and antiquarks - will interact with each

2 Theoretical Framework

other via the exchange of gluons. This is visualised in Figure 2.1 by the black border encircling the six flavours of quarks and the force mediating gluon(s). The study of $SU(3)_C$ interactions is dubbed Quantum Chromodynamics (QCD) based on the Greek word for colour, “chroma”. Even though the concept of colour in this context is by no means related to actually visible colours, the name is quite well chosen due to a peculiar property of QCD called *colour confinement*. It states that solitary quarks cannot exist. Instead, they must unite to form composite, colour neutral (or “white”) particles. Colour neutrality can be achieved in two ways: By “mixing” all of the three colours (anticolours) together or by adding a colour to its corresponding anticolour. The resulting particles are called baryons (antibaryons) and mesons (antimesons)²; they are composite particles consisting of three quarks (antiquarks) and quark-antiquark pairs, respectively.

In contrast to Quantum Electrodynamics (QED) - the description of electromagnetic interactions characterised by the exchange of photons - QCD is non-Abelian. This implies a qualitative difference: whereas photons themselves do not carry any electric charge, gluons carry the colour charge and interact among themselves. The various permutations of colour and anticolour charges constitute the eight gluon colour states represented in the colour octet [3].

Electromagnetism and the weak interaction are two manifestations of the same force that are only distinguishable at low energies. Above the *electroweak unification energy* (≈ 246 GeV [8]), they coalesce into a single force: the electroweak interaction. It is a unified description of electromagnetism and the weak interaction embodied by the expression $SU(2)_L \otimes U(1)_{Y_W}$. Here, the subscript L is used to indicate that $SU(2)_L$ couples only to left-handed fermions (and right-handed antifermions). The physical quantity that governs the behaviour of particles in weak interactions is called weak isospin T_3 . It reverses sign for antiparticles. While all left-handed particles have a weak isospin of $\pm\frac{1}{2}$, all right-handed particles have a weak isospin of zero and therefore do not participate in weak interactions. The generator Y_W of the $U(1)_{Y_W}$ group is called weak hypercharge. It is a function of a particle’s electric charge Q and its weak isospin T_3 :

$$Y_W = 2(Q - T_3) \tag{2.2}$$

Weak isospin T_3 and weak hypercharge Y_W are conserved in all particle interactions. The three gauge bosons of the weak interaction are Z^0 , W^+ and W^- . Unlike all other gauge bosons, the charged W bosons couple to fermion pairs that differ in charge by ± 1 . They possess the unique ability to change the flavour of fermions with which they interact. This property and its ramifications are of great importance. One of the most prominent examples is radioactive decay, as illustrated in Figure 2.2. Since the interaction requires a transfer of electric charge at the vertex, the W boson coupling is said to be associated with a *weak flavour-changing charged current* [3]. Since no charge is transferred in

²The term antimeson can be misleading and is seldom used, since both mesons and antimesons consist of one quark and one antiquark; in an antimeson, quarks are simply replaced by their corresponding antiquarks and vice versa.

2.1 The Standard Model of Particle Physics

interactions that involve the electrically neutral Z^0 boson, the Z^0 is said to be associated with a *neutral current*.

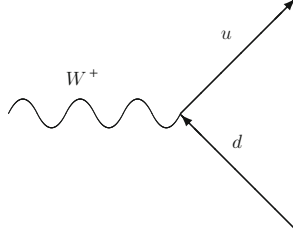


Figure 2.2: Conversion of a down quark into an up quark in nuclear β^+ decay [3].

The $SU(2)_L \otimes U(1)_{Y_W}$ group predicts four massless mediating gauge bosons, whereas experimental values are as high as $80.4 \text{ GeV}/c^2$ for the $W^\pm$ and $91.2 \text{ GeV}/c^2$ for the Z^0 boson [9, 10]. For a long time, this conundrum was puzzling both experimentalists and theorists alike. It was solved by the identification of a process called Spontaneous Symmetry Breaking (SSB) (or “Higgs mechanism”) that allows the full electroweak gauge group $SU(2)_L \otimes U(1)_{Y_W}$ to be hidden away at high energies and replaced by the electromagnetic subgroup $U(1)_{\text{QED}}$ at lower energies, i.e. [11, 12]

$$G_{\text{SM}} = SU(3)_C \otimes SU(2)_L \otimes U(1)_{Y_W} \xrightarrow{\text{SSB}} SU(3)_C \otimes U(1)_{\text{QED}} . \quad (2.3)$$

It can be derived mathematically that electroweak symmetry breaking requires the presence of a complex scalar doublet field $\phi \in \mathbb{C}^2$ (“Higgs field”). An excitation of such a field is detectable as a scalar (spin-0) particle: the aforementioned Higgs-boson. It is the only scalar (spin-0) boson in the SM and furthermore the only SM particle that had not been observed prior to the start-up of the LHC [13].

The fourth fundamental force - gravitation - is not included in the SM and its corresponding mediator - the graviton - has not yet been observed. It is postulated to be a massless spin-2 boson [3, 14]. A summary of the four fundamental interactions is provided in Table 2.1.

Interaction	Mediator	couples to	Strength	Range
Strong	gluon (g)	colour charge C	1	10^{-15} m
Electromagnetism	photon (γ)	electric charge Q	10^{-2}	$\infty (1/r^2)$
Weak	$W^\pm, Z^0$	weak isospin T_3	10^{-5}	10^{-18} m
Gravitation	graviton (G)	mass m	10^{-39}	$\infty (1/r^2)$

Table 2.1: The four fundamental interactions. Relative interaction strengths³ have been normalised with respect to the strong interaction [3].

³The values provided are approximate and serve only as a coarse comparison. The true value of the coupling constant for each interaction depends on the energy scale (length scale) being considered.

In conclusion, the SM is the most widely accepted theory of fundamental particles we have today and its predictions have been verified in numerous experiments. Nevertheless, it is incomplete. One of its major flaws is the lack of gravity. A more fundamental theory is necessary to reconcile Einstein's theory of general relativity with QFT to form a new theory of quantum gravity. Additional problems include the hierarchy problem⁴ and the experimental evidence for neutrino oscillations [15], which implies that neutrinos have non-zero masses [16]. This is in direct contradiction with the predictions of the SM, wherein neutrinos are massless particles propagating at the speed of light. Beyond that, astronomical observations suggest that there may be forms of matter not contained within the SM, most notably dark matter and dark energy [17, 18]. Altogether, this is a strong indication for the existence of physics beyond the SM.

2.2 Beyond the Standard Model Theories

Heavy Z' resonances are predicted in many extensions of the SM, including Grand Unified Theories (GUTs) and theories with extra dimensions. Two very popular models are briefly summarised in this section.

2.2.1 Sequential Standard Model

The simplest extension to the SM is called the Sequential Standard Model (SSM). It involves a spin-1 Z'_{SSM} boson with couplings to fermions equivalent to those of the SM Z boson. The only difference between the SM Z and the Z'_{SSM} is the predicted higher mass of the latter. To incorporate the new Z'_{SSM} boson into the SM gauge group of Equation (2.1), G_{SM} is extended by adding an additional $U(1)'$ group associated with said boson [19]:

$$G_{\text{SM}} = SU(3)_C \otimes SU(2)_L \otimes U(1)_{Y_W} \otimes U(1)' \quad (2.4)$$

While the SSM is poorly motivated from a theoretical perspective [8], it is often considered due to its inherent simplicity and usefulness as a benchmark model [20]. In particular, it allows to compare the relative sensitivity of searches in different channels or from different experiments.

2.2.2 Georgi-Glashow Model

In general, GUTs predict that the SM gauge group G_{SM} is just a subset of a larger symmetry group that reduces to the SM gauge group at lower energies, similar to how the symmetry group $SU(2)_L \otimes U(1)_{Y_W}$ mimics the electromagnetic gauge group $U(1)_{\text{QED}}$ at low energies. Different GUTs can be understood as different guesses as to what the larger symmetry group might look like. Many Grand Unification Models are based on the E_6 symmetry group described in [21]. One of these is the *Georgi-Glashow Model*

⁴the fact that the energy scale of electroweak symmetry breaking, $\mathcal{O}(1 \text{ TeV})$, is much smaller than the gravitational Planck scale, $M_{\text{Pl}} \sim \mathcal{O}(10^{19} \text{ GeV})$

2.3 Feynman Diagrams

[11, 22], in which strong, electromagnetic and weak forces arise from a single fundamental interaction based on an $SU(5)$ Lie group. It implies that the SM gauge group G_{SM} is the result of spontaneous symmetry breaking of E_6 via $SU(5)$ that is predicted to occur below some high energy scale known as the *grand unification scale* ($\sim \mathcal{O}(10^{14} \text{ GeV})$ [8]):

$$\begin{aligned} E_6 &\rightarrow SU(5) \otimes U(1)_\chi \otimes U(1)_\psi \\ &\rightarrow SU(3)_C \otimes SU(2)_L \otimes U(1)_{Y_W} \otimes U(1)' \\ &\rightarrow SU(3)_C \otimes SU(2)_L \otimes U(1)_{Y_W} \\ &\rightarrow SU(3)_C \otimes U(1)_{\text{QED}} \end{aligned} \quad (2.5)$$

In this model, the E_6 gauge group is broken into an $SU(5)$ and two additional $U(1)$ gauge groups named $U(1)_\chi$ and $U(1)_\psi$. The lightest linear combination of the corresponding two new neutral gauge bosons, Z'_ψ and Z'_χ , is considered the Z' candidate, i.e. [23]

$$Z'(\theta_{E_6}) = Z'_\psi \cos \theta_{E_6} + Z'_\chi \sin \theta_{E_6} \quad (2.6)$$

where $0 \leq \theta_{E_6} < \pi$ is the mixing angle between the two gauge bosons. In order to be able to observe the new $Z'(\theta_{E_6})$ boson at the LHC, the symmetry breaking of $SU(5)$ has to occur in two steps as shown in Equation (2.5), with the breaking of $U(1)'$ involving a Vacuum Expectation Value (VEV) at the TeV scale [8].

Unlike the SSM, GUT models are physically motivated by the fact that gauge couplings of the SM interactions seem to converge towards the same value at very high energy scales.

2.3 Feynman Diagrams

Particle interactions can be represented by so-called Feynman diagrams⁵. Examples are shown in Figure 2.3. The convention used here is to have the time axes point upward and the space axes from left to right. Feynman diagrams play a crucial role in calculating measurable quantities such as cross sections, from which decay rates and probabilities for particular particle interactions to occur can be derived. Every vertex of a diagram is associated with a mathematical term in the calculation, contributing a factor proportional to $\sqrt{\alpha}$ to $\mathcal{M}$ [3], where α is the coupling constant representing the strength of the interaction and $|\mathcal{M}|^2$ the Feynman invariant amplitude that allows to compute observable quantities. For example, the differential cross section $d\sigma$ of a two body scattering process is given by [8]

$$d\sigma = \frac{|\mathcal{M}_{\beta\alpha}|^2}{4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}} (2\pi)^4 \delta^4(k_\beta - p_\alpha) d\beta \quad , \quad d\beta = \prod_{f \in \beta} \frac{d^3 \mathbf{k}_f}{2E_f (2\pi)^3} \quad (2.7)$$

where the subscripts 1 and 2 refer to the initial state (i.e. incoming) particles, $p_{1/2}$ denote their four-momenta and p_α (k_β) is the sum of the four-momenta of the initial (final) state

⁵named after theorist Richard Feynman

2 Theoretical Framework

particles, and $\mathbf{k}_f$ is the three-momentum of the final state particles. Calculations keeping only the first term in the series expansion for $\mathcal{M}$ are referred to as Leading Order (LO) calculations. Those keeping higher order terms in the couplings are called calculations at Next-to-Leading Order (NLO), Next-to-Next-to-Leading Order (NNLO), and so on.

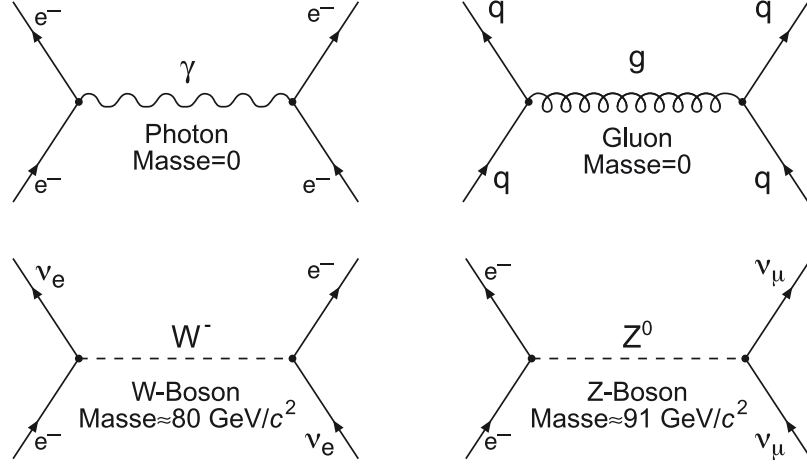


Figure 2.3: Feynman diagrams representing the interaction of two particles [5].

Chapter 3

Experimental Setup

This chapter presents an overview of the ATLAS experiment at the LHC located at CERN near Geneva, Switzerland. Section one introduces the LHC and its accelerator chain. Section two focuses on the primary constituent parts of the ATLAS detector. For a more detailed description of individual accelerator components and/or detector elements, consult [24, 25].

3.1 The Large Hadron Collider

With an intended centre-of-mass energy of $\sqrt{s} = 13$ TeV in the first data taking period of Run II and a design luminosity of $10^{34} \text{ cm}^{-2}\text{s}^{-1}$ [25], the CERN Large Hadron Collider (LHC) is currently the most powerful high energy particle collider in the world. It is the most advanced experimental tool for probing SM predictions, and its potential for discoveries indicative of new physics is far beyond any of its predecessor experiments.

Beam pipes in the LHC are kept at ultrahigh vacuum to avoid particle losses due to collisions with gas molecules. Strong electromagnets¹ are deployed to direct the particle beams around the accelerator rings. Several quadrupoles are positioned at different stages of the accelerator chain to ensure proton bunches remain in a tight beam. All magnets in the LHC ring operate in a superconducting state, i.e. electricity is conducted without losses attributable to electrical resistance. This is necessary to reach the high magnetic fields required to curve particles' paths at velocities greater than 99.9% the speed of light. To maintain their superconducting state, magnets are kept at a temperature of 1.9 K. To reach this temperature, the LHC uses a cryogenic system to disseminate superfluid helium, which exhibits an exceptionally high thermal conductivity.

The proton beam passes through a series of accelerators before it is injected into the 27 km long LHC ring. Each accelerator boosts the energy of the beam before passing it

¹magnets whose magnetic field is produced by the flow of electric current

3 Experimental Setup

on to the next in the sequence. The proton source is a cylinder filled with compressed hydrogen gas. Hydrogen atoms are ionised through interactions with free electrons in the duoplasmatron, leaving behind bare, positively charged protons. These are then directed into the accelerator chain by means of an electric field. The accelerator chain is illustrated in Figure 3.1.

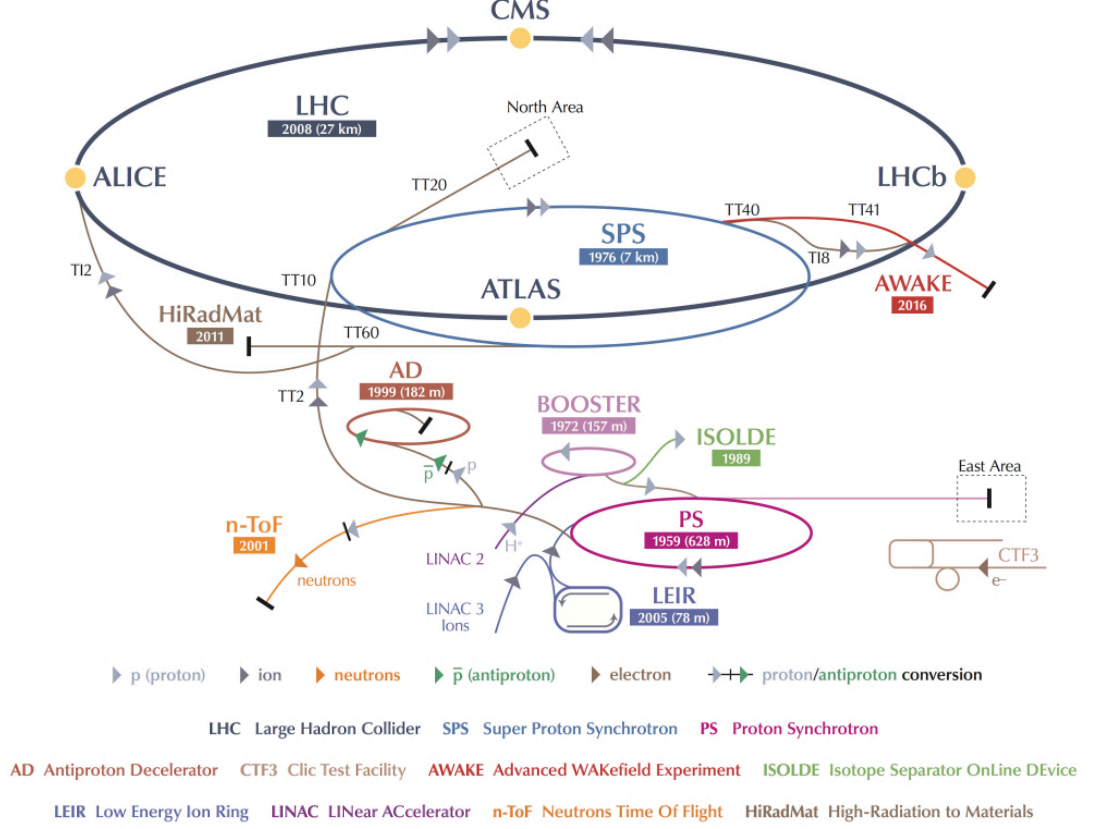


Figure 3.1: The CERN accelerator complex [26].

The injector chain for the ATLAS and CMS experiments is LINAC 2 - PSB - PS - SPS - LHC. The first accelerator in the chain is the **Linear Accelerator 2** (LINAC 2). It uses Radiofrequency (RF) cavities to generate alternating electric potentials, effectively “pushing” protons from behind and “pulling” them from the front. Cavities are set out in increasing length with respect to the propagation of the beam, causing protons arriving later (earlier) than those in phase with the electric field to be accelerated more (less) while taking into account the particles’ increasing velocities. When protons enter the Proton Synchrotron Booster (PSB), they have reached an energy of 50 MeV. At this point, straight acceleration becomes impracticable. Consequently, all subsequent pre-accelerators are synchrotrons. The PSB consists of four superimposed synchrotron rings. To further accelerate the circulating protons, a pulsed electric field is applied, repeatedly “pushing” the particles every time they pass a certain point in the ring. In order to keep

the circumference of the particle trajectories constant while the velocity increases, the force exerted by the surrounding electromagnets has to increase consistently. The PSB accelerates protons to an energy of 1.4 GeV. They are then forwarded to the Proton Synchrotron (PS), where their energy reaches 25 GeV before once again being passed on to the Super Proton Synchrotron (SPS), a larger synchrotron that allows for even higher beam energies. As a general rule, the maximum particle energy that is attainable in a synchrotron is limited by the synchrotron’s circumference and the magnetic field strength provided by the magnets: The more powerful the magnets, the more particles’ trajectories can be bent, i.e. a small ring size will suffice if strong magnets are available. However, to reach the highly relativistic velocities required in the proton-proton collisions of ATLAS and CMS, both a large ring and very powerful magnets are needed. Finally, at an energy of 450 GeV, the proton bunches are injected into the LHC ring. Table 3.1 gives an overview of the synchrotrons employed in the injector chain of ATLAS and CMS. The LHC is included for comparison.

Accelerator	C [m]	$E_{\max}$ [GeV]
PSB	157	1.4
PS	628	25
SPS	7.000	450
LHC (Run Ia)	27.000	3.500
LHC (Run Ib)	27.000	4.000
LHC (Run II)	27.000	6.500

Table 3.1: Circumference C and maximum attainable beam energy $E_{\max}$ in pre-accelerator synchrotrons and the LHC

Once circulating in the LHC ring, the counter-rotating beams cross over in four detector caverns, where they can be made to collide. The four main LHC detectors installed to track the debris of the collisions are ALICE, ATLAS, CMS and LHCb. A Large Ion Collider Experiment (ALICE) records data from heavy-ion (i.e. lead ions) collisions in order to study the formation of quark-gluon plasma, a phenomenon crucial to our understanding of colour confinement. Large Hadron Collider beauty (LHCb) focuses on measuring the parameters of CP violation in interactions involving bottom quarks². Findings in this experiment could potentially explain the baryon asymmetry problem, i.e. the imbalance of matter and antimatter within the observable universe. ATLAS and CMS are general-purpose detectors used to investigate a large range of physics. Even though both experiments pursue the same scientific goals, they are led independently and their detectors use different technical solutions. For instance, they exhibit a different magnet-system design. This approach reduces experimental bias, at the same time allowing for cross-confirmation of new discoveries, as has been the case in the discovery of a Higgs-like boson in 2012 [1, 2].

²In this context often referred to as b or “beauty” quarks.

3 Experimental Setup

The analysis in the following chapter uses data collected by the ATLAS experiment. Accordingly, characteristic features of the ATLAS detector are discussed in the next section.

3.2 The ATLAS Detector

ATLAS - **A Toroidal LHC Apparatus** - is one of two general-purpose detectors at the LHC. It is aligned concentrically around the beam pipe and exhibits a cylindrical symmetry. With dimensions of 44m x 25m (height x diameter) it is one of the largest volume particle detectors that have ever been built. Figure 3.2 gives an overview of the ATLAS detector. As can be seen, it comprises a toroidal magnet system and multiple subsystems sensitive to one or more specific types of particles that emerge from the collisions in the LHC. With the exception of neutrinos, every SM particle passing through the detector will yield a signal (if detector inefficiencies are ignored). Figure 3.3 illustrates what particle species different layers of the detector are sensitive to. The remainder of this chapter is aimed at outlining the operating principles of the most important detector components. A description of the coordinate system that is used throughout this section is provided in Appendix A.

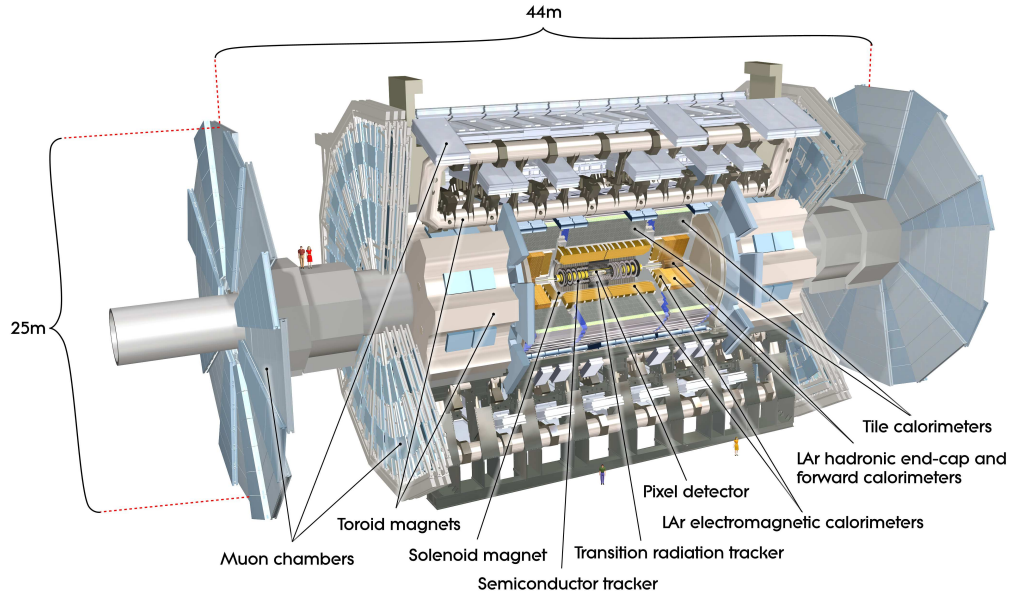


Figure 3.2: Cut-away view of the ATLAS detector [25].

3.2.1 Inner Detector

The Inner Detector (ID) is designed to measure the trajectories of charged particles. It is composed of the Pixel Detector, the **S**emiconductor **T**racker (SCT) and the Transition

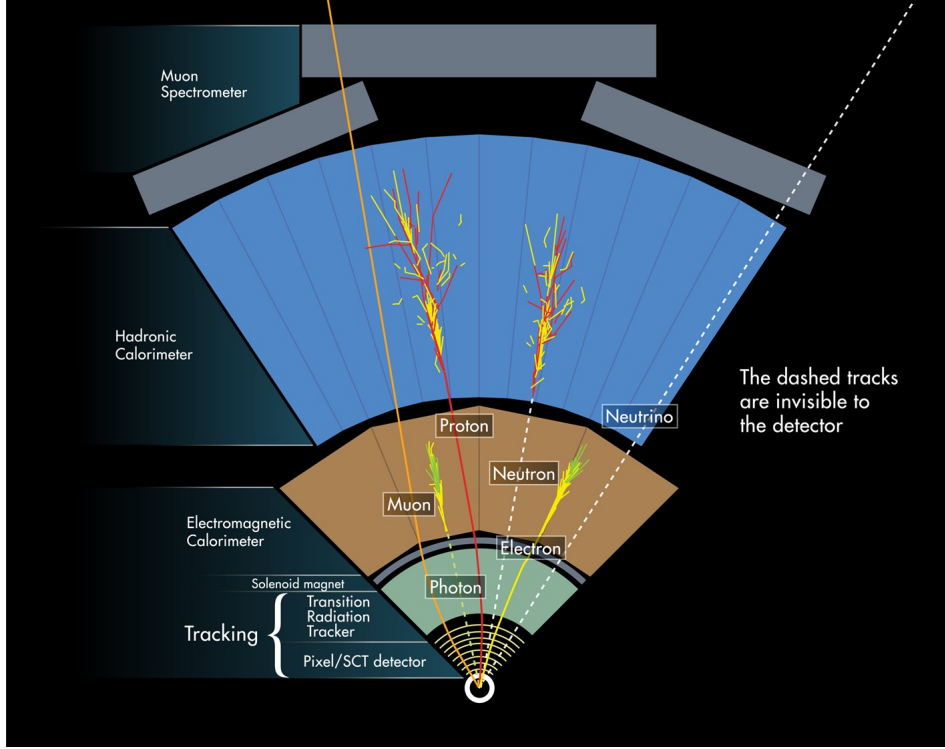


Figure 3.3: Particle detection in the ATLAS detector (schematic) [27].

Radiation Tracker (TRT) and is immersed in a 2 T magnetic field generated by the central solenoid. Pixel Detector and SCT cover the pseudorapidity range $|\eta| < 2.5$, while the TRT covers $|\eta| < 2.0$ [25]. Each of them consists of several barrel layers, which are centred around the interaction point, and disks or end-cap wheels perpendicular to the beam axis at either end of the barrel. This is illustrated in ???. Closest to the interaction point is the Pixel Detector.

3.2.1.1 Pixel Detector

The Pixel Detector uses silicon pixel sensors to record particle tracks. When a charged particle traverses one of the silicon modules, it ionises atoms in the lattice, causing the formation of electron-hole pairs. Measuring the resulting current allows to reconstruct particle tracks. This is essential to locate primary and secondary vertices [28], thus making it possible to discriminate between collision points. The transverse momentum p_T of charged particles can be determined based on their curvature in the solenoidal field. In addition, the direction of curvature will reveal a particle's charge.

Performing high-precision measurements despite the large track density produced by collisions in the LHC poses a challenge. Very fine detector granularity is needed to arrive at good results in these extreme conditions. To cope with this difficult task,

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the Pixel Detector is furnished with more than 80 million individual readout channels, comprising three cylindrical layers in the barrel region and three disks in each end-cap. To further improve the detector's vertex reconstruction capabilities, an additional Insertable B-Layer (IBL) was installed between the existing Pixel Detector and a new smaller radius beam pipe during the first shutdown period of the LHC machine [29, 30].

3.2.1.2 Semiconductor Tracker

The SCT relies on the same functional principle as the Pixel Detector, but is made of silicon microstrips instead of silicon pixels. It consists of four double layers in the barrel region and nine double-layered disks in each end-cap. Consecutive layers are tilted by an angle of 40 mrad with respect to one another, enabling three dimensional reconstruction of particle trajectories. In total, the SCT contains approximately 6 million readout channels.

3.2.1.3 Transition Radiation Tracker

Unlike Pixel Detector and SCT, the TRT is constructed of straw drift tubes containing a xenon-based gas mixture (Xe-CO₂-O₂). They are aligned parallel to the beam axis in the barrel region and perpendicular to the beam axis in the end-cap regions (see ??). A gold-plated W-Re (tungsten-rhenium) wire is located at the centre of each tube. While the wires are kept at ground potential, the walls of the tubes are operated at -1350 V [25]. When a charged particle passes through one of the straw drift tubes, it will ionise gas molecules contained within the tube. Driven by the electric tension the liberated electrons will drift towards the W-Re wire (anode) at the centre of the tube, whereas gas ions will accumulate at the wall (cathode). The separation of charges can be recorded and translated into a signal.

Characteristic feature of the TRT is the ability to differentiate electrons from more massive mesonic (and hadronic) particles of equal charge, such as negatively charged pions or kaons. This is made possible by means of Transition Radiation (TR), a phenomenon that occurs as a charged particle moves across the interface of two media with different dielectric constants [31]. In the TRT, this is achieved by interleaving layers of straws with polypropylene fibres [32] (so-called “radiators”). The energy loss of a charged particle due to TR is proportional to the particle's Lorentz factor γ ($\gamma = E/mc^2$). The TRT's readout system differentiates two signal thresholds: A low threshold that is triggered by ionisation, and a high threshold triggered by the much stronger signal in the presence of TR, thus allowing to distinguish electrons from heavier particles of equal charge.

3.2.2 Calorimetry

Surrounding the ID is the calorimeter system, namely the Electromagnetic Calorimeter (ECal), the Hadronic Calorimeter (HCal) and the Forward Calorimeter (FCal) (see Figure 3.4). Its primary objective is to measure the energy and position of electrons, photons (ECal and FCal) and hadrons (HCal and FCal), and to provide containment

for electromagnetic and hadronic particle showers, i.e. preventing these particles from reaching the Muon Spectrometer (MS) [13]. Unlike magnetic spectrometers, calorimeters are also sensitive to neutral particles, e.g. neutral mesons or neutrons. Based on their distinct interactions with the detector, it is possible to distinguish electrons and photons from pions and muons, for example. Moreover, they allow for indirect detection of neutrinos through measurement of an event’s missing energy. Calorimeters deployed in the ATLAS detector are sampling calorimeters with ϕ -symmetry and complete and uniform coverage around the beam axis [25]. A sampling calorimeter is composed of alternating layers of a dense absorber material to absorb incident particles and an active material to produce an output signal that is proportional to the energy of the incident particle³ [34]. Table 3.2 summarises the η coverage of individual calorimeter components. Combined, ECal, HCal and FCal cover the pseudorapidity range $|\eta| < 4.9$ [25]. An overview of the calorimeter system is shown in Figure 3.4. For brevity, the FCal will not be discussed in this paper.

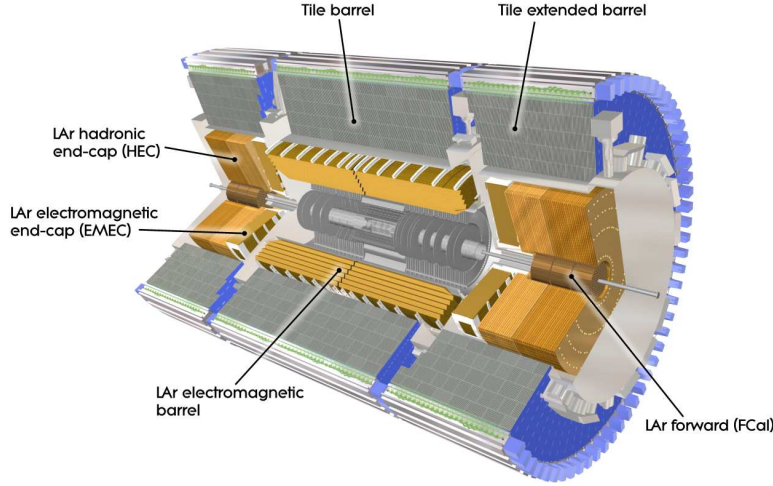


Figure 3.4: Cut-away view of the ATLAS calorimeter system [25].

3.2.2.1 Electromagnetic Calorimeter

The ECal uses lead absorber plates and Liquid Argon (LAr) as an active medium. It comprises the LAr electromagnetic barrel and two LAr Electromagnetic End-Caps (EMECs). Above 1 GeV, electrons and photons produce secondary photons via bremsstrahlung and secondary electrons and positrons via pair production in photon-photon collisions ($\gamma + \gamma \rightarrow e^+ + e^-$), respectively. The result is a cascade of particles (“particle shower”) with progressively diminishing energies (illustrated in Figure 3.3). At lower energies,

³In contrast, homogeneous calorimeters are built of only one type of material that performs both tasks. A comprehensive summary of different calorimeter types - including their assets and drawbacks - is given in [33].

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Calorimeter Element	η Coverage	Rad. Lengths [X_0]	Int. Lengths [λ]
LAr ECal barrel	$ \eta < 1.475$	> 22	-
LAr EMECs	$1.375 < \eta < 3.2$	> 24	-
HCal tile barrel	$ \eta < 1.0$	-	≈ 9.7
HCal extended tile barrel	$0.8 < \eta < 1.7$	-	≈ 9.7
LAr HECs	$1.5 < \eta < 3.2$	-	≈ 10

Table 3.2: Pseudorapidity coverage and total thickness in terms of radiation lengths (X_0) or interaction lengths (λ) of the ATLAS calorimeters [25].

particles from the shower ionise atoms in the LAr layers. The ions can then be collected and converted into a signal that is proportional to the energy of the incident particle.

An accordion geometry is used for the absorbers and electrodes of the ECal barrel and end-caps to reduce detector response time. A signal response of 50 ns or less is needed to meet the requirements for operation at the LHC, where on average ... protons collide per ... [33]. In a classical calorimeter, the ionisation signal produced by particle showers is collected by electrodes located in the middle of the LAr layers, whereas in the accordion-shaped ECal, the signal is extracted directly from the front and back faces of the calorimeter. This minimises the amount of capacitance and inductance inducing cables, thereby reducing the overall response time. See Figure 3.5 for a comparison of ordinary and accordion calorimeter geometry.

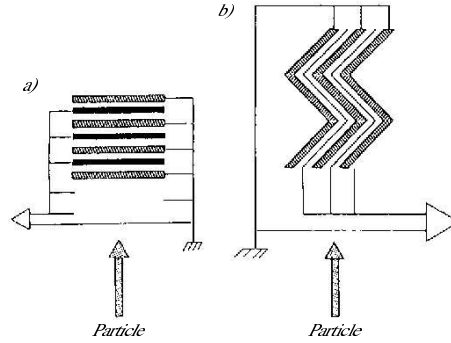


Figure 3.5: Traditional a) and accordion b) calorimeter geometry [33].

3.2.2.2 Hadronic Calorimeter

The HCal is designed specifically to measure the energy of hadrons. It consists of one tile barrel, two extended tile barrels and two LAr Hadronic End-Caps (HECs). Analogous to the EMECs, the HECs are LAr sampling calorimeters using copper plates as absorber material. They are each composed of two consecutive wheels placed orthogonal to the beam axis with an absorber thickness of 25 and 50 mm, respectively [35]. The hadronic

tile calorimeters are sampling calorimeters using scintillating tiles as the active material and iron as the absorber. The scintillating tiles are made of ..., a material known to show fluorescence. When struck by a charged particle or high-energy photon, the ... absorb its energy and re-emit it in the form of photons (a process called scintillation). In contrast to LAr calorimeters, where the signal yield is proportional to the amount of charge collected by the electrodes, the signal strength picked up by the tiles is proportional to the amount of light, i.e. the number of photons and their respective energies. A calorimeter of this type is referred to as scintillator calorimeter. Both sides of the scintillating tiles are read out by wavelength shifting fibres into two separate photomultipliers [36].

3.2.3 Muon Spectrometer

The MS forms the outermost layer of the ATLAS detector. It is used for the identification, triggering and high-precision measurement of muon momenta. Momentum measurements are based on the deflection of muon tracks in the magnetic field provided by superconducting air-core toroid magnets. More specifically, one large barrel toroid for the range $|\eta| < 1.4$ and two smaller end-cap toroids for $1.6 < |\eta| < 2.7$ [25], as depicted in Figure 3.6.

The four chamber systems that make up the MS can be subdivided according to their intended purpose into precision measurement tracking chambers (MDTs and CSCs) and trigger chambers (RPCs and TGCs). They are briefly introduced in the remainder of this subsection. Their respective positions within the MS are indicated in Figure 3.6. A quantitative overview is presented in Table 3.3. Chambers are rectangular in the barrel region where they form three cylindrical layers (“stations”) concentric with the beam axis. In the end-caps, chambers are trapezoidal and placed vertically, also in three stations. They are labelled I (“inner”), M (“middle”) and O (“outer”) as determined by their position with respect to the interaction point. A cross-section of the barrel stations is shown in Figure 3.7. The naming convention for chambers is as follows: The first letter specifies the detector region (“B” for barrel or “E” for end-cap), followed by the station and lastly the chamber size (“L” for large or “S” for small). Exceptions for the last label are made for chambers in the vicinity of the detector’s feet and rail system. They are designed specifically to incorporate the additional supporting material and exhibit variant dimensional characteristics.

3.2.3.1 Monitored Drift Tubes

Monitored Drift Tubes (MDTs) are aluminium tubes of 30 mm diameter that operate very similar to the TRT’s straw drift tubes. Muons traversing a tube ionise an argon-based gas mixture (Ar-CO₂-H₂O), causing an electron drift towards the central W-Re anode wire. Due to the larger dimension and smaller CO₂ concentration compared to the straw drift tubes, the maximum electron drift time in the MDTs is about 700 ns [25]. This is much longer than the electron drift time in the straw drift tubes (50 ns), yet sufficient enough because track rates are significantly lower in the MS than in the

3 Experimental Setup

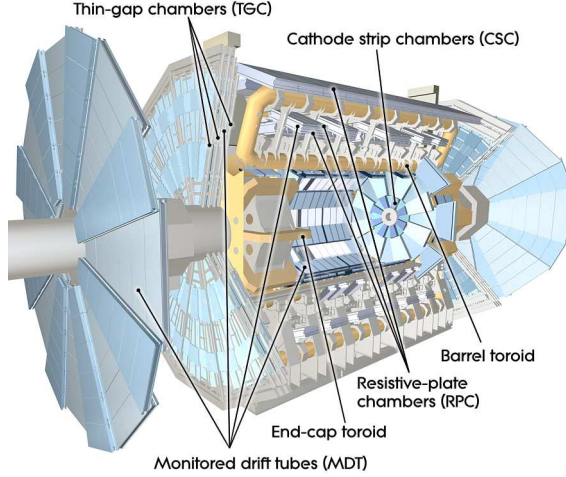


Figure 3.6: Cut-away view of the ATLAS muon system [25].

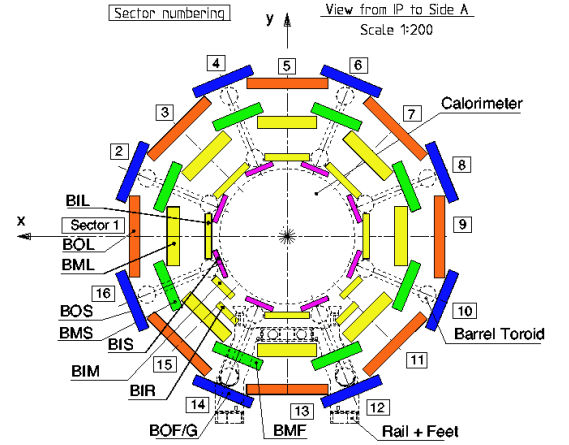


Figure 3.7: Cross-section of the barrel muon system [25].

ID, in consequence of the shielding provided by the calorimeter system [8].

MDT chambers are constructed from 2x4 monolayers of drift tubes for the inner station and 2x3 monolayers for the middle and outer stations [36]. This is done to improve muon track resolution and to increase redundancy for pattern recognition. Approximately 12,000 internal optical sensors are installed to monitor alignment and potential mechanical deformation of MDTs due to temperature changes [25]. While MDTs provide measurements over most of the η -range, they can only operate up to about 150 Hz/cm² [37]. This limit is exceeded for $|\eta| > 2.0$. To obtain results at larger pseudorapidities, Cathode Strip Chambers (CSCs) are used.

3.2.3.2 Cathode Strip Chambers

CSCs are multi-wire proportional chambers with segmented cathode readout. Cathode strips providing the precision coordinate η are orthogonal to the anode wires, while those providing the transverse coordinate ϕ are parallel. Every CSC consists of four identical layers, each with 192 precision and 48 transverse coordinate strips [37]. This is illustrated in Figure 3.8. The position determination relies on the interpolation of the charge induced on the segmented cathode by the electron avalanche created on the anode wire. CSCs feature an excellent spatial resolution ($\leq 70 \mu\text{m}$ per layer in the precision coordinate at a rate of 1 kHz/cm²), a short electron drift time ($< 45 \text{ ns}$) and the capability to operate at high counting rates (up to ...) [37]. The gas mixture used in the CSCs is simply Ar-CO₂. The lack of hydrogen and the small volume occupied by the gas ensure low sensitivity to neutrons. To gain coverage in the range $2.0 < |\eta| < 2.7$, 16 CSCs are deployed close to the interaction point in the innermost station of each end-cap.

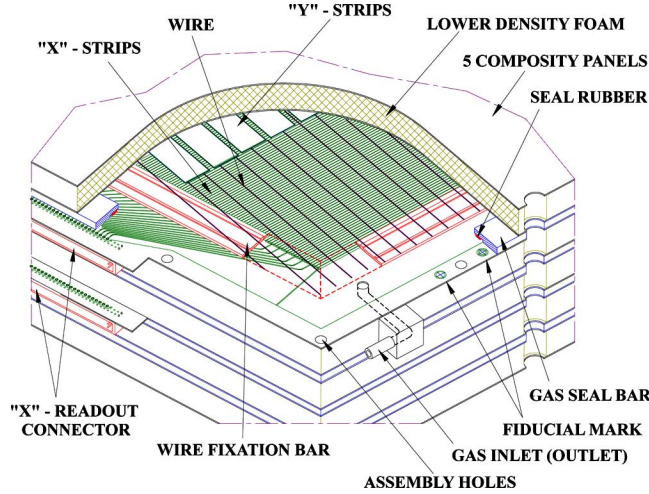


Figure 3.8: Internal arrangement of the anode wires and the cathode strips in a CSC. X-strips measure the precision coordinate η , Y-strips the transverse coordinate ϕ [37].

3.2.3.3 Resistive Plate Chambers

Resistive Plate Chambers (RPCs) are used primarily as a first-level muon trigger in the barrel region ($|\eta| < 1.05$), but also to provide a complementary second coordinate measurement in the non-bending ϕ -projection [25]. They are assembled together with MDTs of equal dimensions in a common mechanical support structure [38]. Similar to the CSCs, an RPC's readout strips are grouped in two (η and ϕ) panels orthogonal to each other. Every chamber includes two gaseous detectors, each containing two parallel resistive electrode plates separated by a gap of 2 mm that is occupied by the gas ($\text{C}_2\text{H}_2\text{F}_4$ - C_4H_{10} - SF_6). Insulating polycarbonate spacers are placed every 10 cm to make sure the distance between electrode plates is kept constant. A strong uniform electric field is applied by the resistive electrodes, resulting in an avalanche of electrons with a drift time of only ... ns. Via capacitive coupling, a signal is induced on both sides of the gap. It is read out by copper strips before being transmitted to the front-end electronics. The time resolution achieved by an RPC is ... ns, making them very well suited for muon triggering. The layout of an RPC can be seen in Figure 3.9.

3.2.3.4 Thin Gap Chambers

Muon triggering in the end-cap regions ($1.05 < |\eta| < 2.4$) is provided by Thin Gap Chambers (TGCs). They operate on the same principle as multi-wire proportional chambers, allowing also for an additional measurement of the azimuthal ϕ coordinate in the bending (i.e. radial) direction. TGC layers are constructed in doublets and triplets. The inner station consists of one doublet and is used solely for the determination of the second coordinate. The remaining seven layers are positioned before (one triplet) and after (two doublets) the outermost MDT layer. Muon triggering in the end-cap regions

3 Experimental Setup

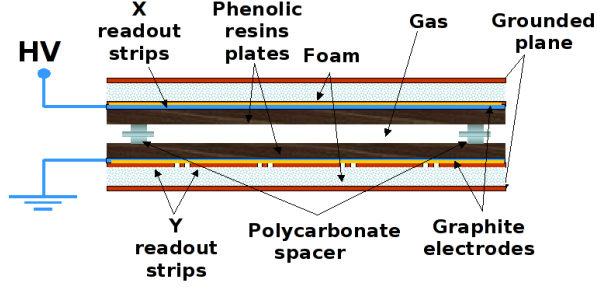


Figure 3.9: Structure of an RPC [25].

is very challenging for various reasons: Since muon momenta, corresponding to a given p_T , increase significantly with η , the η -dependent detector granularity in the end-caps has to be increased in order to obtain a similar p_T resolution as in the barrel region. This becomes even more apparent due to the fact that the three outer layers of TGCs are located outside of the magnetic field, i.e. they cannot detect any curvature, and their respective distances are smaller compared to the RPC layers in the barrel. Another difficulty arises from magnetic field inhomogeneities in the transition region. This is discussed in more detail in what follows.

Type	# chambers	η Cov.	X Res.	Wire Diameter	Op. Voltage
MDT	1150	$ \eta < 2.0$	80 μm	50 μm	3.08 kV
CSC	32	$2.0 < \eta < 2.7$	60 μm	30 μm	1.9 kV
RPC	606	$ \eta < 1.05$	-	-	9.8 kV
TGC	3588	$1.05 < \eta < 2.4$	-	50 μm	2.9 ± 0.1 kV

Table 3.3: Comparison of operating and performance parameters of the four chamber technologies employed in the Muon Spectrometer of ATLAS [25].

3.2.4 Magnet System

The magnet system of ATLAS is one of its most distinctive features [8]. Its configuration is shown in Figure 3.10. It consists of a central solenoid (pink) to provide curvature in the ID and three toroidal magnets - one barrel toroid (red) and two end-cap toroids (rose) - within the MS.

Each of the toroids is composed of eight coils arranged symmetrically around the z -axis. Their air-core design reduces the amount of material traversed by muons in an attempt to prevent the deterioration of resolution due to multiple scattering. Even though the field generated by the central solenoid does not interfere with the toroidal fields, there is a transition region ($1.4 < |\eta| < 1.6$) where the magnetic fields of the barrel and end-cap toroids overlap [25, 36]. In this range, the magnetic field integral becomes small and even briefly changes sign (see Figure 3.11), making momentum measurements extremely difficult [8]. All magnets are operated at a temperature of 4.5 K.

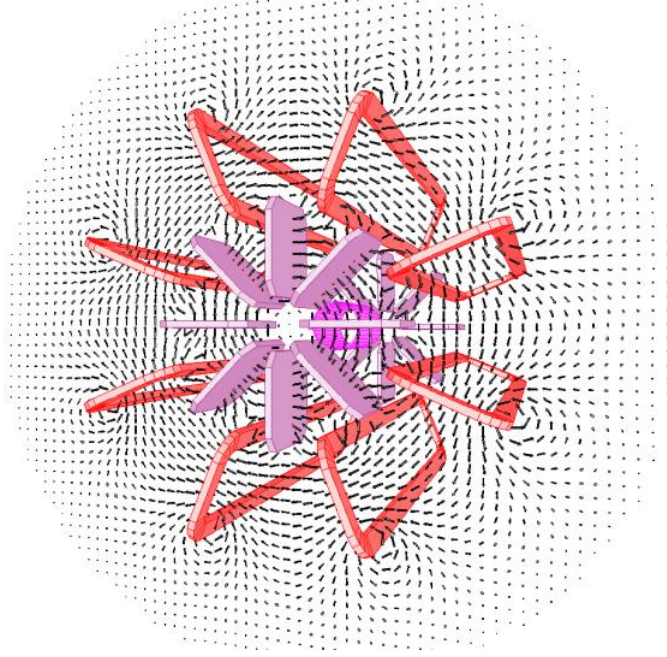


Figure 3.10: Visual representation of the magnetic field generated by the ATLAS magnet system [39].

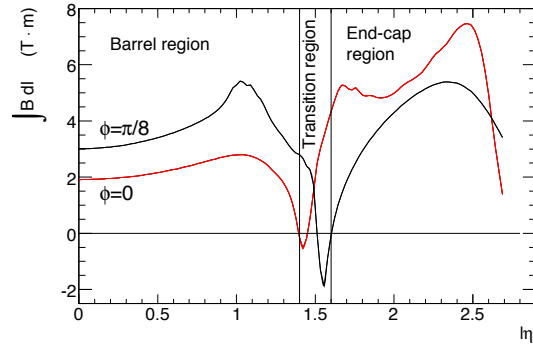


Figure 3.11: ATLAS Muon Spectrometer integrated magnetic field strength as a function of $|\eta|$ for $\phi = 0$ (red line) and $\phi = \pi/8$ (black line) [40].

Chapter 4

Data Analysis

This analysis focuses on the search for a high-mass Z' resonance in the decay mode $pp \rightarrow Z' + X \rightarrow \ell^+\ell^- + X$, where $\ell^+\ell^-$ corresponds to either a dielectron (e^+e^-) or dimuon ($\mu^+\mu^-$) final state. If such a resonance exists, it will manifest itself through its decay as a peak in the dilepton invariant mass ($m_{\ell\ell}$) distribution. Per definition [10], the invariant mass is given by

$$m_{\text{inv}} = \left[(E_1 + E_2)^2 - (\vec{p}_1 + \vec{p}_2)^2 \right]^{1/2} \quad (4.1)$$

and can be determined solely from the energies and momenta of the decay products, which are labelled 1 and 2. Using ATLAS coordinates (cf. Appendix A), it can be expressed as

$$m_{\text{inv}} = \left[m_1^2 + m_2^2 + 2(E_T(1)E_T(2)\cosh(\Delta y) - \vec{p}_T(1) \cdot \vec{p}_T(2)) \right]^{1/2} \quad (4.2)$$

where

$$E_T(i) = \sqrt{|\vec{p}_T(i)|^2 + m_i^2} \quad i = 1, 2 \quad (4.3)$$

Dilepton resonances are considered to be among the most promising channels in the search for new physics due their fully reconstructable signal and well-understood backgrounds [20]. After discussing event selection techniques and backgrounds, this chapter elaborates on the relevance of Parton Distribution Functions. This is followed by a few concise remarks on systematic uncertainties. Finally, signal and background modelling procedures are presented and the results from limit calculations are displayed.

4.1 Event Selection

All candidate events must satisfy data-quality requirements, i.e. proper functioning of the ID, MS, solenoid and toroid magnets, calorimeter and trigger subsystems, as well

as stable LHC beams [23]. To be considered in the search for high-mass resonances decaying to dielectron or dimuon final states, collision events must meet a predefined set of (channel dependent) identification criteria.

For the dielectron (e^+e^-) channel, at least two reconstructed electron candidates within $|\eta| < 2.47$ are required. In each event, the electron candidate with the higher E_T (“leading electron”) and the electron candidate with the lower E_T (“subleading electron”) must satisfy $E_T > 40$ GeV and $E_T > 30$ GeV, respectively [20]. An opposite charge requirement is not placed on electron candidates because of possible charge misidentification, either due to bremsstrahlung or to the limited momentum resolution of the ID at very high transverse momentum p_T . Electron candidates consist of clusters of energy deposited in the ECal that can be traced back to ID particle tracks. Shower profiles produced in the calorimeter system must be consistent with those expected for electromagnetic showers and exceed a minimum requirement on the amount of TR (cf. Section 3.2.1.3, closing paragraph). The pseudorapidity range $1.37 \leq |\eta| \leq 1.52$ is excluded as it corresponds to a transition region between the barrel and end-cap calorimeters which exhibits degraded energy resolution [20, 23]. Background originating from photon pair production (cf. Section 3.2.2.1, opening paragraph) can be suppressed by demanding a hit in the first layer of the pixel detector. To reduce background from jets misidentified as electrons, the leading electron is required to be isolated such that [23]

$$\sum E_T(\Delta R < 0.2) < 0.007E_T + 5.0 \text{ GeV} \quad (4.4)$$

where $\sum E_T(\Delta R < 0.2)$ is the sum of transverse energies around the electron direction in a cone of $\Delta R < 0.2$. Similarly, the subleading electron is required to satisfy

$$\sum E_T(\Delta R < 0.2) < 0.022E_T + 6.0 \text{ GeV}. \quad (4.5)$$

The difference in the isolation requirement takes into account the different energy losses due to bremsstrahlung. If more than one electron pair is found in an event, the one with the highest transverse momentum scalar sum of the two electrons is selected.

In the dimuon ($\mu^+\mu^-$) channel, candidates must satisfy $p_T > 25$ GeV and $|\eta| < 2.4$ [23]. The transverse momentum is obtained from a combined fit to the measurements of the ID and the MS. Muon pairs must be composed of two oppositely charged muons, otherwise they are discarded. Each muon is required to have a minimum number of hits in each of the ID components. Muons that pass the selection criteria are classified as either three-station or two-station muons, depending on the number of hits in each station of the MS. Three-station muons have at least three hits in every MS station and at least one hit in two layers of the trigger chambers (RPCs and TGCs). These muons have a p_T resolution ranging from 10% to 25% at 1 TeV [23]. In the end-cap regions of the MS, the hit requirement on the inner station corresponds to at least two hits in the CSCs. Two-station muons are selected from the central region ($|\eta| < 1.05$) and must provide at least five hits in both the inner and outer station of the MS and at

least one hit in one layer of the trigger chambers. Their p_T resolution is slightly worse compared to three-station muons. To suppress background from cosmic rays, muons are also required to satisfy selections on the transverse impact parameter, $|d_0| < 0.2$ mm, and the z -coordinate with respect to the primary vertex (PV), $|z_0 - z(\text{PV})| < 1$ mm [23]. Analogous to the dielectron channel, background from jets can be reduced by imposing an isolation requirement on each muon, namely [23]

$$\frac{\sum p_T(\Delta R < 0.3)}{p_T(\mu)} < 0.05 \quad (4.6)$$

where $p_T(\mu)$ is the transverse momentum of the muon and $\sum p_T(\Delta R < 0.3)$ is the sum of transverse momenta of all other tracks in a cone of $\Delta R < 0.3$ around the direction of the muon. If a muon pair comprises two three-station muons, the event is said to pass the primary or “tight” dimuon selection. This category constitutes about 90% of the sample. If no tight dimuon candidate is found, pairs can be built with one three-station and one two-station muon. Such events are said to pass the secondary or “loose” dimuon selection. In either instance, the dimuon candidate with the highest transverse momentum scalar sum is chosen if more than one candidate is found in an event. Only muon pairs that pass the primary dimuon selection are considered in Section 4.5 and Section 4.6.

4.2 Backgrounds

For both channels, the dominant and irreducible¹ background stems from the Drell-Yan process ($q\bar{q} \rightarrow Z/\gamma^* \rightarrow \ell^+\ell^-$) [41, 42]. It occurs when a quark and an antiquark² annihilate to form a Z bosons or a virtual photon (denoted γ^*), which ultimately decays into a dilepton final state. The corresponding Feynman diagrams are shown in Figure 4.1.

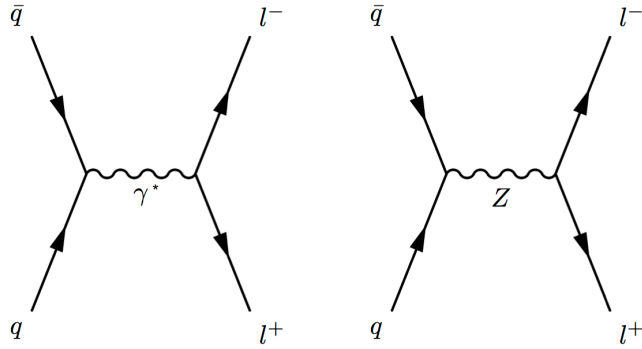


Figure 4.1: Leading Order contributions to the Drell-Yan process [43].

¹“Irreducible” means that it is impossible to define a variable that allows to distinguish background events from signal events.

²The presence of antiquarks in proton-proton collisions will become clear in Section 4.3.

Figure 4.3 shows all background contributions for the dielectron (top) and dimuon (bottom) channel at $\sqrt{s} = 8$ TeV. Two Z'_{SSM} signal line shapes corresponding to two different mass hypothesis (1.50 and 2.50 TeV) are also shown. They are generated using signal templates (cf. Figure 4.7 - Figure 4.10).

In the dielectron channel, the next largest background originates from heavy flavour quarks, multijet and $\gamma + \text{jet}$ production (referred to hereafter as dijet background) as well as $W + \text{jet}$ processes and top quark production with $W + \text{jets}$ (referred to hereafter as $W + \text{jet}$ background) [20, 23]. Additional contributions are diboson (WW , ZZ , WZ) production and the dileptonic decay of $t\bar{t}$ pairs, as illustrated in Figure 4.2³. Since precisely modelling the probability of a jet to be misidentified as an electron (“fake rate”) is difficult, background contributions from events with jets or photons in the final state that pass all channel selection criteria are determined as a function of E_T and η using background-enriched data samples. All other signal and background expectations are evaluated with simulated samples using the highest order available cross section predictions.

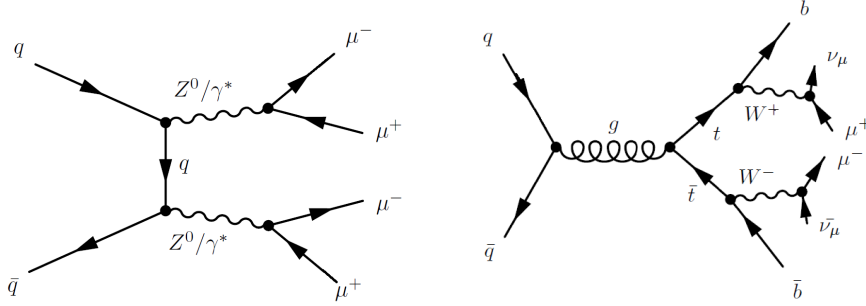


Figure 4.2: Diboson (left) and $t\bar{t}$ (right) production [44].

In the dimuon channel, the Drell-Yan process is followed by (in order of dominance) $t\bar{t}$ and diboson production. Contributions from dijet and $W + \text{jet}$ backgrounds are negligible, as is the background caused by cosmic rays. Cosmic muons are produced when high-energy protons from astronomical sources collide with molecules in the Earth’s atmosphere. The outcome of such a collision is a cascade of hadronic particles. Among these particles are mesons, which often decay into muons. Due to relativistic time dilation, the muons have long lifetimes, allowing them to traverse the Earth’s surface and pass through the ATLAS detector, thus potentially (if coincident with an LHC produced collision event) contaminating the recorded data.

For the analysis performed in this thesis, only the Drell-Yan background is taken into account at LO. Since all other background contributions are smaller by roughly one order of magnitude for most of the dilepton invariant mass range (cf. Figure 4.3), this

³Note that in this figure the axes for time and space have been interchanged with respect to the convention introduced in Section 2.3.

4 Data Analysis

does not diminish the accuracy of the results significantly.

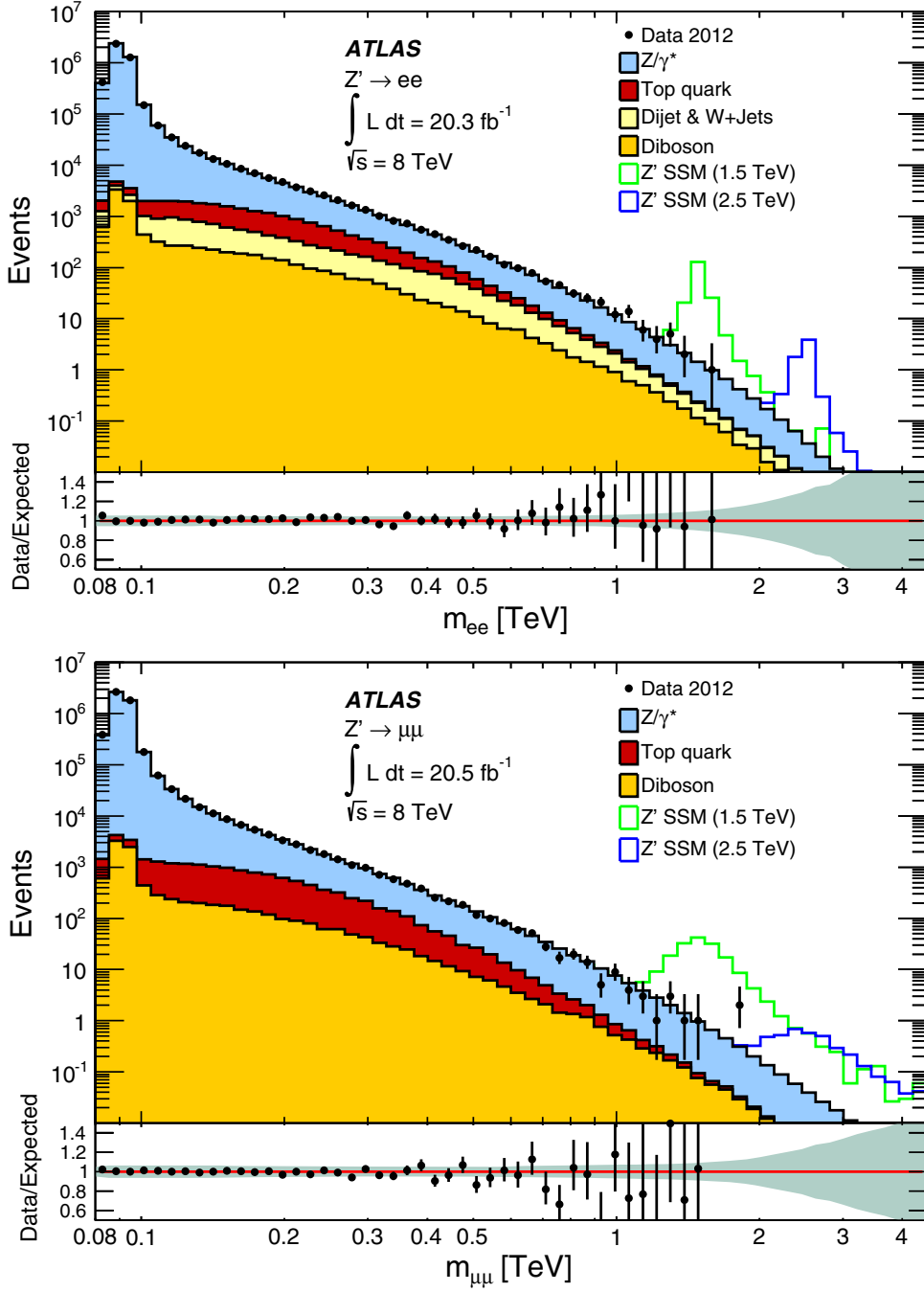


Figure 4.3: Dielectron (top) and dimuon (bottom) invariant mass distributions after event selection, with two selected Z'_{SSM} signals overlaid, compared to the stacked sum of all expected backgrounds. The bin width is constant in $\log m_{\ell\ell}$. The ratios of data and background are displayed below. The green band in the ratio plot shows systematic uncertainties [20].

4.3 Parton Distribution Functions

The notion of protons being made up of three quarks (uud) is an oversimplification that holds true only in the low energy regime. In actuality, a proton consists of *valence quarks* - the quarks that determine its quantum numbers - which are embedded in a sea of virtual quark-antiquark pairs generated by the gluons⁴ holding the quarks together. Altogether, the constituents of the proton are referred to as “partons”. Parton Distribution Functions (PDFs) are momentum distribution functions of the partons within the proton. They represent probability densities $f_i(x, Q^2)$ to find a parton of flavour i (quarks or gluon) carrying a momentum fraction x of the proton momentum at a squared energy scale Q^2 [45]. The knowledge of proton PDFs is inferred from scattering processes, mainly the Deep Inelastic Scattering (DIS) of leptons off nucleons, whereby the resolving power increases with the energy of the probe (i.e. the lepton). Figure 4.4 shows two examples of PDFs computed at NLO for Q^2 values of 10 GeV^2 (left) and 10^4 GeV^2 (right). Note that in this case, the probability density function has been normalised with respect to the number of partons. As can be seen, the amount of proton momentum that is not carried by valence quarks (u and d) increases at higher energies as more and more quark-antiquark pairs start to take on a low momentum fraction x . Also, the momentum fraction carried by gluons increases with increasing Q^2 .

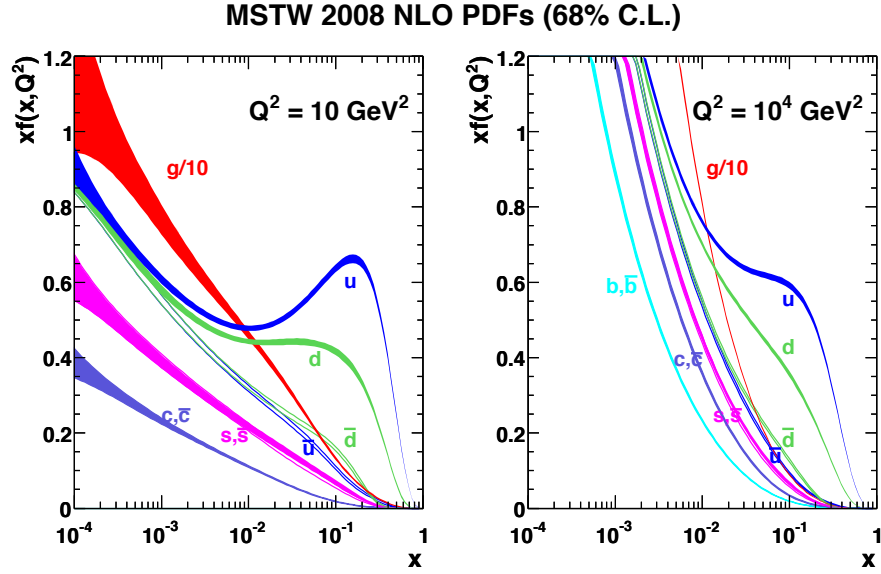


Figure 4.4: Parton Distribution Functions for $Q^2 = 10 \text{ GeV}^2$ (left) and $Q^2 = 10^4 \text{ GeV}^2$ (right) computed at NLO. Coloured contours indicate a 68% (1σ) CL [46].

PDFs are a fundamental ingredient in essentially all forms of theoretical predictions of

⁴This is another ramification of the non-Abelian nature of QCD. In principle, it is similar to charge screening in QED, albeit resulting in an opposite effect: Probing a bare colour charge will yield smaller values at shorter distances (corresponding to higher energies), until interactions between partons become arbitrarily weak. This phenomenon is known as *asymptotic freedom*.

SM and BSM processes at hadron colliders. The Z' search is no exception. In fact, the variations of PDFs have a large impact on the differential Z/γ^* cross section and the Z' signal cross section as a function of the dilepton invariant mass $m_{\ell\ell}$. For simplicity, only the latter is considered in this analysis using a set of PDFs provided by the MSTW (Martin-Stirling-Thorne-Watt) collaboration [46] varied by their systematic uncertainties. These uncertainties are parametrised using mutually independent parameters, which are called “eigenvectors” of the PDF set in function space [8]. They can be varied in orthogonal directions to propagate the systematic uncertainties associated with PDF variations to calculated physical quantities. The MSTW2008LO PDF set for the Z' analysis contains 41 eigenvectors and provides all information necessary to perform LO calculations. The relative uncertainties of PDF eigenvectors are shown in Figure 4.5 for eighth different values of the Z' pole mass m_p . For every eigenvector, the Z' cross section is computed as a function of mass using output data from 100,000 simulated events that are created and processed by the event generator PYTHIA. The asymmetric uncertainties are calculated using the following equations [8]:

$$\Delta\sigma^- = \sqrt{\sum_{i=1}^n (\max(\sigma_0 - \sigma_i^+, \sigma_0 - \sigma_i^-, 0))^2} \quad (4.7)$$

$$\Delta\sigma^+ = \sqrt{\sum_{i=1}^n (\max(\sigma_i^+ - \sigma_0, \sigma_i^- - \sigma_0, 0))^2} \quad (4.8)$$

where n is the number of PDF eigenvectors, σ_0 is the cross section for the central value PDF, σ_i^- is the cross section for the lower value of the i^{th} PDF eigenvector and σ_i^+ is the cross section for the higher value of the i^{th} PDF eigenvector. The resulting uncertainties are summarised in Table 4.1 for twelve mass points ranging from 0.50 to 3.75 TeV.

Z' Mass [TeV]	Uncertainties [%]	
0.50	- 2.53719	+ 3.89808
1.00	- 3.41904	+ 5.69168
1.50	- 6.42154	+ 4.16177
1.75	- 6.11771	+ 5.53384
2.00	- 6.87700	+ 5.34541
2.25	- 8.11271	+ 5.85815
2.50	- 9.04256	+ 6.55335
2.75	- 9.61754	+ 7.52411
3.00	- 10.7541	+ 7.16023
3.25	- 11.5592	+ 7.56618
3.50	- 11.2084	+ 9.27046
3.75	- 12.4119	+ 8.15492

Table 4.1: Uncertainty on the Z' signal cross-sections due to PDF variations at $\sqrt{s} = 13$ TeV computed at a 90% CL using MSTW2008LO.

4 Data Analysis

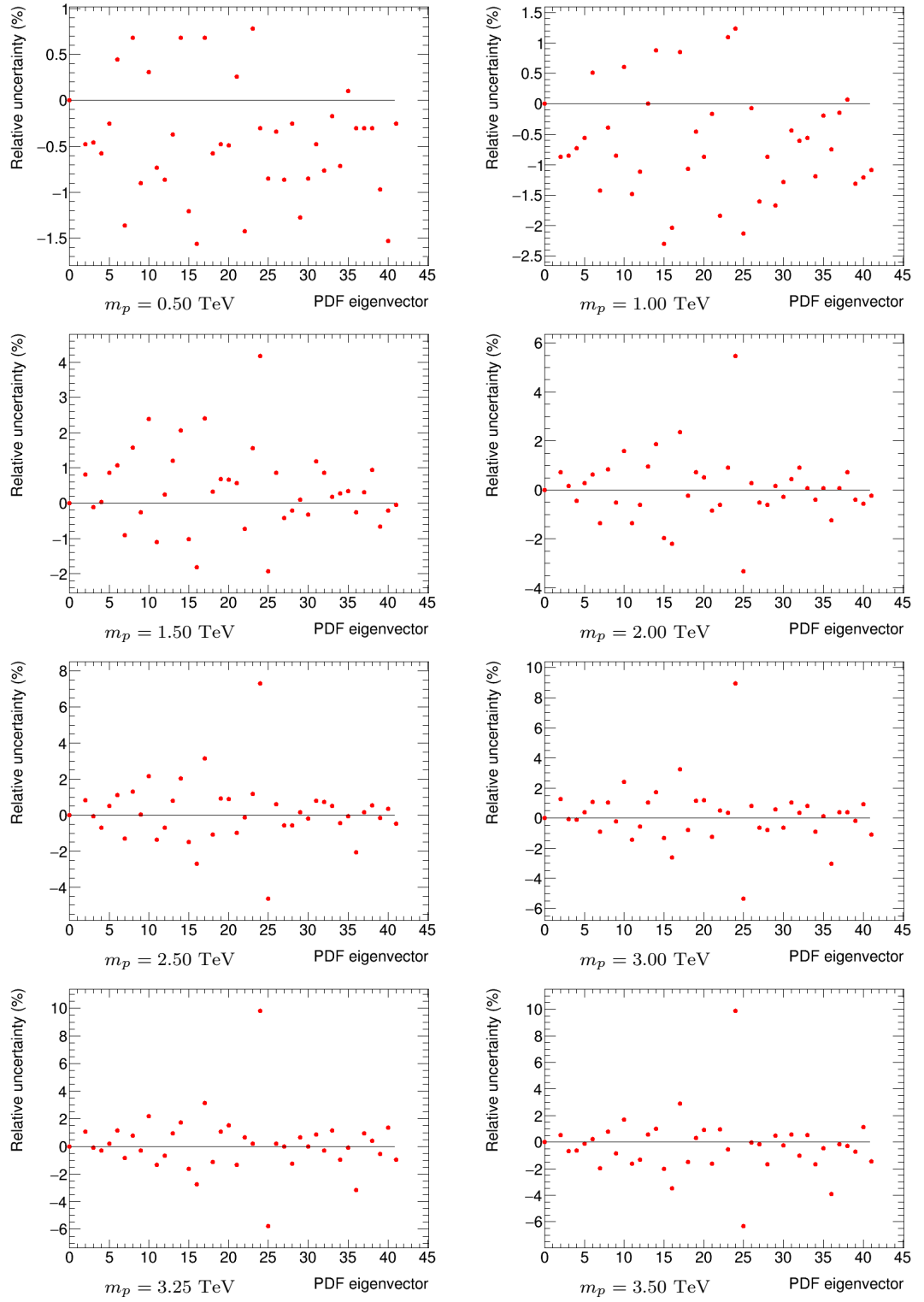


Figure 4.5: Relative uncertainties of PDF eigenvectors for different pole masses m_p .

4.4 Systematic Uncertainties

To simplify the treatment of systematic uncertainties in this analysis, the background contribution from the Drell-Yan process is normalised to the data region of the Z boson peak (80 - 110 GeV)⁵. The advantage of this approach is that mass-independent uncertainties, such as the uncertainty on the integrated luminosity $\mathcal{L}_{\text{int}}$, will cancel in the statistical analysis between event yields in the high-mass and the normalisation region. Therefore, only mass-dependent systematic uncertainties on the background prediction need to be considered. To account for the uncertainty on the Z/γ^* cross section in the normalisation region, a mass-independent systematic error of 5% is assigned to the signal expectation [23]. The Z/γ^* cross section values used in this normalisation approach are obtained from theoretical calculations at NNLO. They are $\sigma_{Z/\gamma^*} = 1147.4$ pb at $\sqrt{s} = 8$ TeV and $\sigma_{Z/\gamma^*} = 1970.6$ pb at $\sqrt{s} = 13$ TeV for $m_{\ell\ell} > 80$ GeV.

The dominating mass-dependent systematic uncertainties are theoretical uncertainties originating from PDFs as well as QCD and electroweak corrections. Because PDFs are not known to high accuracy at large momentum fractions x , uncertainties on the parton luminosity become relatively large as the kinematic limit is approached [20]. Experimental effects, such as, for instance, muon trigger and reconstruction efficiency, are also taken into account. They are assumed to be fully correlated between signal and all types of backgrounds. A comprehensive summary listing all uncertainties considered in the Z' search is provided in [20, 23]. All systematic uncertainties estimated to have an impact of $\leq 3\%$ on the expected number of events are presumed to have a negligible impact on the limit setting and are consequently neglected in the statistical analysis.

4.5 Signal and Background Modelling

To conduct the search for a high-mass Z' boson, the dilepton invariant mass line shape is examined for a positive deviation of the observed event yields corresponding to a new resonance. Signal and background expectations are typically represented using ROOT histograms. The expected yield of events is provided by background and signal templates and is given in bins of $m_{\ell\ell}$. The observed invariant mass distributions $m_{\mu\mu}$ and m_{ee} are then compared to the expectation from SM backgrounds. In order to be able to detect a significant excess of events that is indicative of new physics, the expected number of events from SM backgrounds must be known to very high precision, especially in the presence of irreducible backgrounds. Since no excess of events is found in the data recorded by ATLAS in Run I of the LHC, exclusion limits on the mass of a Z' boson are set at a 95% Confidence Level (CL).

⁵As previously mentioned, only the Drell-Yan contribution is taken into account in this analysis. For a more general and precise analysis, the sum of Drell-Yan, $t\bar{t}$ and diboson background contributions would have to be considered in the normalisation procedure.

4.5.1 Signal and Background Templates

The main background contributions in the Z' analysis (discussed in Section 4.2) are evaluated using simulated samples inferred from Monte Carlo (MC) simulations. This involves generating events for the physical processes of interest and simulating the interactions with a virtual representation of the ATLAS detector. The expected number of collision events for given scenario is [8]

$$N_{\text{exp}} = \mathcal{L}_{\text{int}} \sum_i (\sigma B A \varepsilon)_i \quad (4.9)$$

where $\mathcal{L}_{\text{int}} = \int \mathcal{L}(t) dt$ is the integrated luminosity, σ_i is the cross section and B_i the branching ratio of a given physical process i into final states of interest, A_i is the acceptance of the detector for this process and ε_i is the efficiency, i.e. the probability for an accepted event to pass the analysis selection criteria. Mathematically, the likelihood for two events to occur one after another is given by the product of their probabilities (assuming the events are independent of each other). Thus, the product of acceptance A and efficiency ε can be expressed as the fraction of simulated events that are accepted by the virtual ATLAS detector AND ($\wedge$) fulfil all of the selection requirements imposed by the analysis, labelled N_{pass} , over the total number of generated events N_{tot} , i.e.

$$A\varepsilon = \frac{N_{\text{pass}}}{N_{\text{tot}}} . \quad (4.10)$$

As pointed out in Section 4.2, background contributions originating from jets or photons are evaluated using data-driven techniques, but turn out to be negligible.

In order to be able to evaluate the significance of differences between the observed data and the expected backgrounds, signal templates are needed in addition to background templates. For each signal hypothesis, the combined signal and background templates are compared to the data using a likelihood fit [8]. The template shape approach has a decisive advantage: It uses the full shape of the distribution (as opposed to a simple counting method using only an invariant mass window). This makes the analysis robust against uncertainties in the background model at high mass and reduces the likelihood of uncertainties in the background estimate to mask a potential signal. Also, it allows for a better sensitivity in the case of a signal with a low mass tail arising from off-shell production, which occurs due to the steeply falling PDF of the two colliding protons at high momentum transfer [20], as illustrated in Figure 4.6. This low mass tail is commonly referred to as *parton luminosity tail*. Its size increases with the resonance width, and its impact on the $m_{\ell\ell}$ distribution grows as the kinematic limit is approached. Since the resonance width depends on the particle's lifetime and coupling strength to SM fermions, the parton luminosity tail is larger for wider resonances, and the limits are worse as compared to those obtained from narrower resonances. Parton luminosity tails are visible in signal templates of high mass hypotheses, such as the samples provided in Figure 4.7 (dimuon template samples) and Figure 4.9 (dielectron template samples).

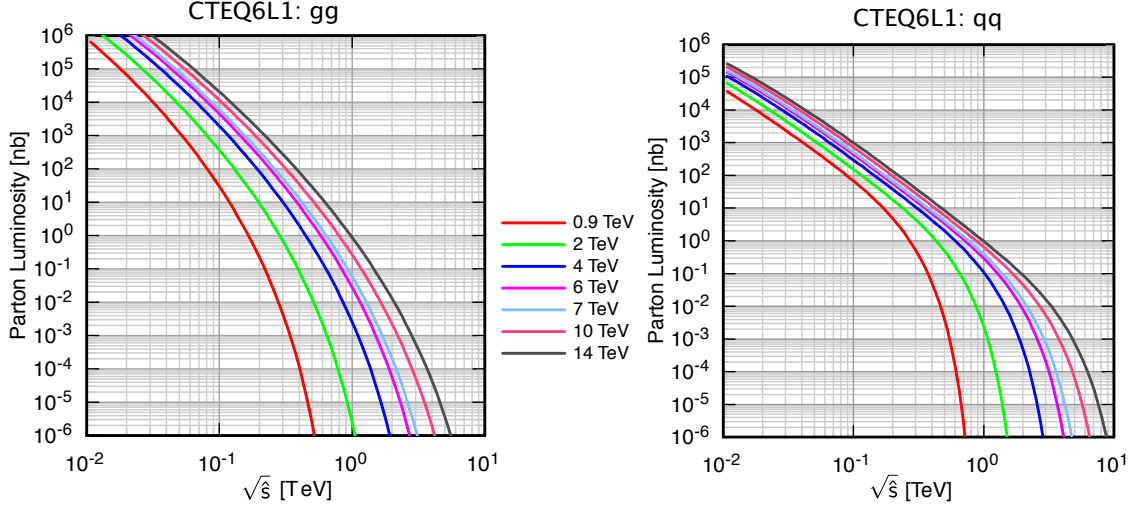


Figure 4.6: Parton luminosities for gg (left) and qq (right) interactions at various centre-of-mass energies [47].

4.5.2 Acceptance $\times$ Efficiency

The results of ATLAS and CMS in the search for dilepton resonances cannot be compared directly, since deviating assumptions are made about the Z' signal shape. Whereas the CMS assumption is based on a Breit-Wigner function convoluted with a Gaussian resolution function, ATLAS assumes a MC based Z' shape which incorporates a Breit-Wigner factor (consult [48] for details on the CMS assumption, [49] for details on the ATLAS assumption, and [50] for a comparison). In an attempt to allow for a better comparison between results of the two collaborations, all limit calculations in this thesis are performed twice: One time using the original template files provided by ATLAS, and a second time with modified templates that are meant to mimic the signal shape assumption of the CMS collaboration. The original signal templates are shown for two different mass hypotheses in Figure 4.7 (dimuon channel) and Figure 4.9 (dielectron channel), and the corresponding modified templates in Figure 4.8 (dimuon channel) and Figure 4.10 (dielectron channel). Note that empty bins are not displayed in the modified muon template samples (cf. Figure 4.8 and Figure 4.10). This can be traced back to a slight inconsistency in the coding of the ROOT macros that controls how output files are saved. It is merely an optical disparity and does not affect any of the calculations in the analysis.

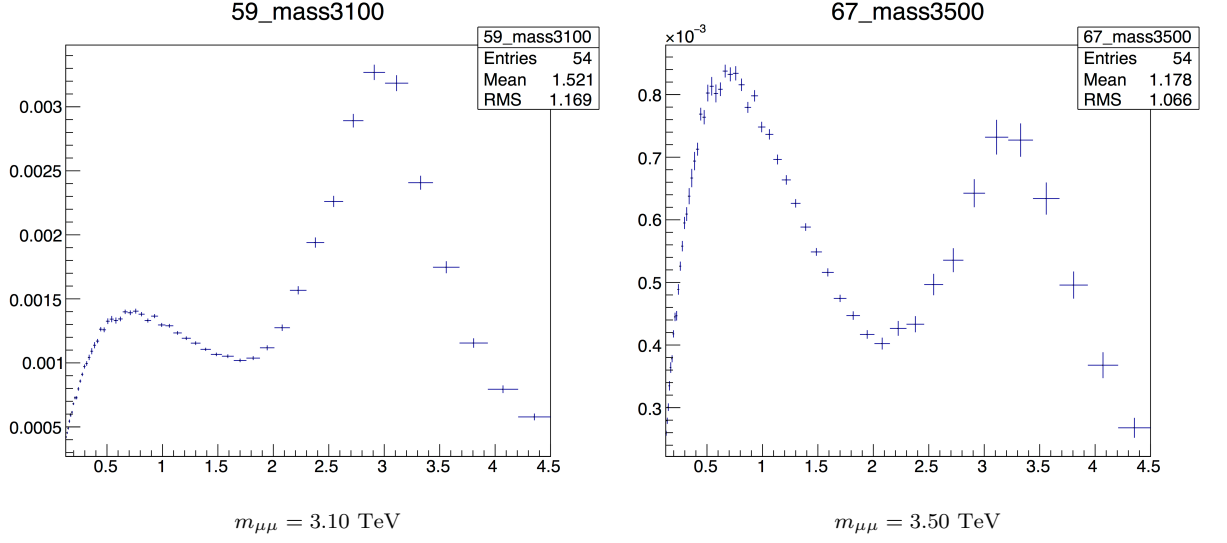


Figure 4.7: Original muon templates for $m_{\mu\mu} = 3.10$ TeV (left) and $m_{\mu\mu} = 3.50$ TeV (right).

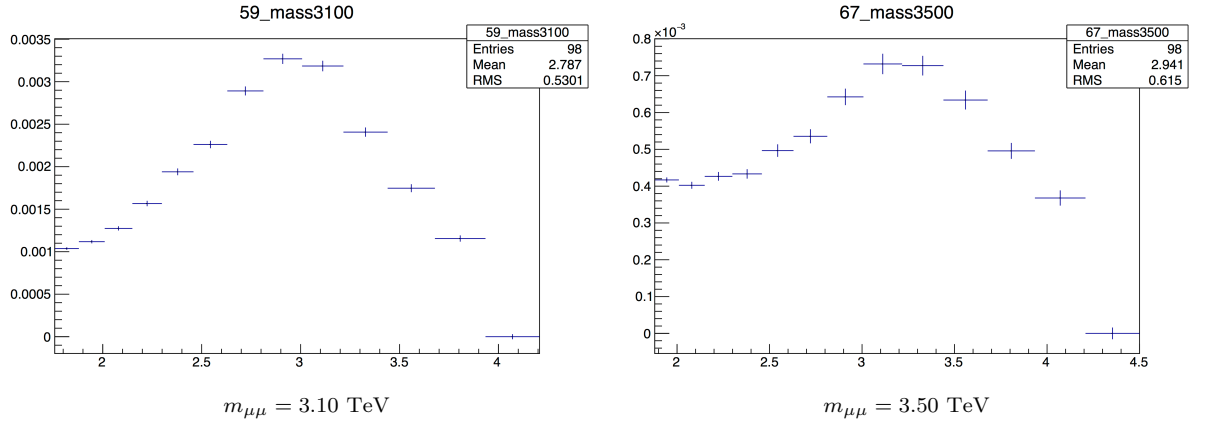


Figure 4.8: Modified muon templates for $m_{\mu\mu} = 3.10$ TeV (left) and $m_{\mu\mu} = 3.50$ TeV (right).

As can be seen, the parton luminosity tail is cut off in the modified templates. As one would expect, the effect of the parton luminosity tail is visible in the limit plots of ATLAS, but not in any of the plots issued by CMS. This is illustrated by means of a comparison of the two combined limit plots presented in Figure 4.11.

As has been derived in Section 4.5.1, the product of acceptance and efficiency is related to the fraction of simulated candidate events that are accepted by the virtual ATLAS representation and pass the dilepton event selection requirements over the total number of simulated events. Figure 4.12 shows $A \times \varepsilon$ as a function of the Z'_{SMM} pole mass. Due

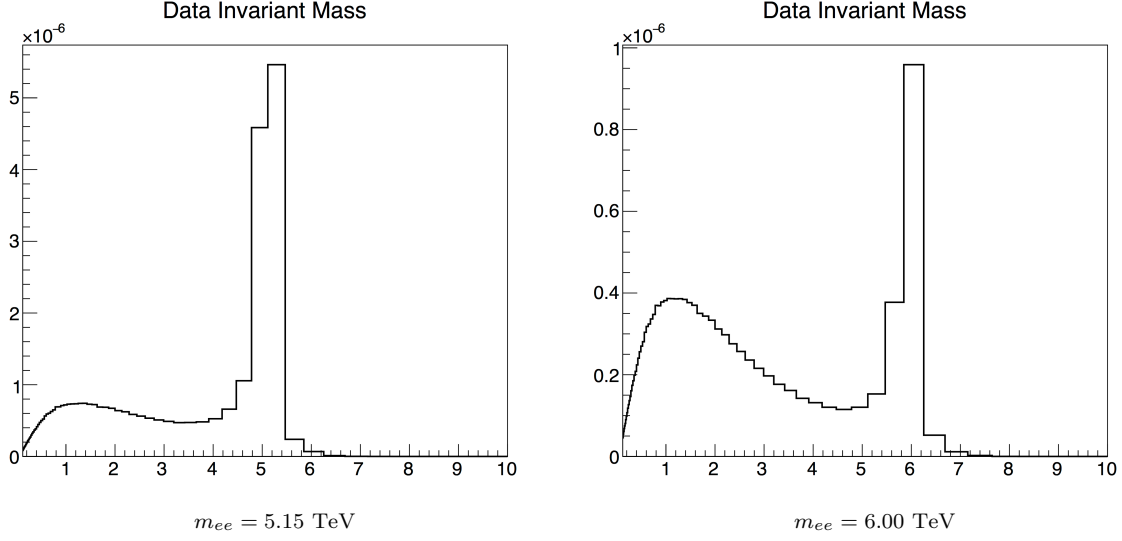


Figure 4.9: Original electron templates for $m_{ee} = 5.15$ TeV (left) and $m_{ee} = 6.00$ TeV (right).

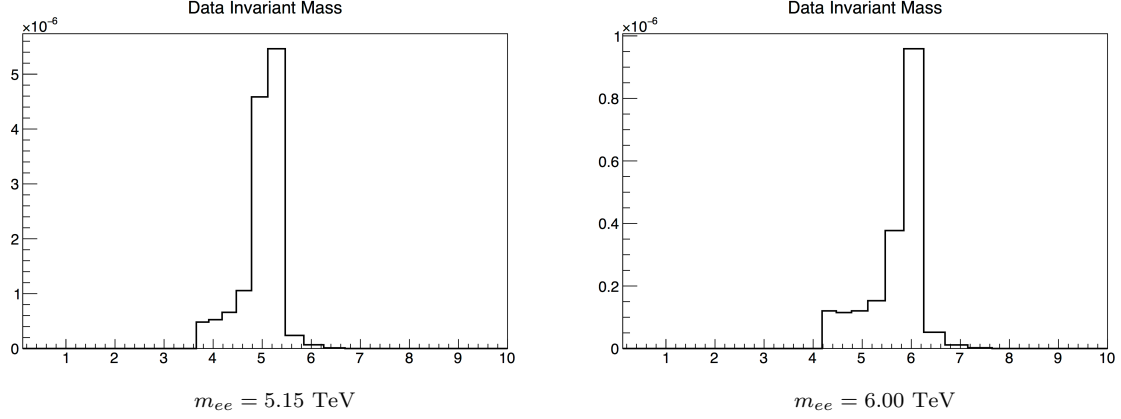


Figure 4.10: Modified electron templates for $m_{ee} = 5.15$ TeV (left) and $m_{ee} = 6.00$ TeV (right).

to kinematic selection requirements, $A \times \varepsilon$ rises for low values of the Z'_{SSM} pole mass. It decreases at high values because the strong decrease of the parton luminosity at high momentum transfer enhances the relative fraction of events in the low mass tail of the spectrum (as can be seen in the original template samples) arising from off-shell Z'_{SSM} production [20].

The modification of the signal templates reduces the number of signal events being considered in the analysis. This effect has to be taken into account by also modifying the acceptance times efficiency fits that are used in the limit calculations. This is done by rearranging the familiar Equation (4.10) so as to eliminate the total number of events

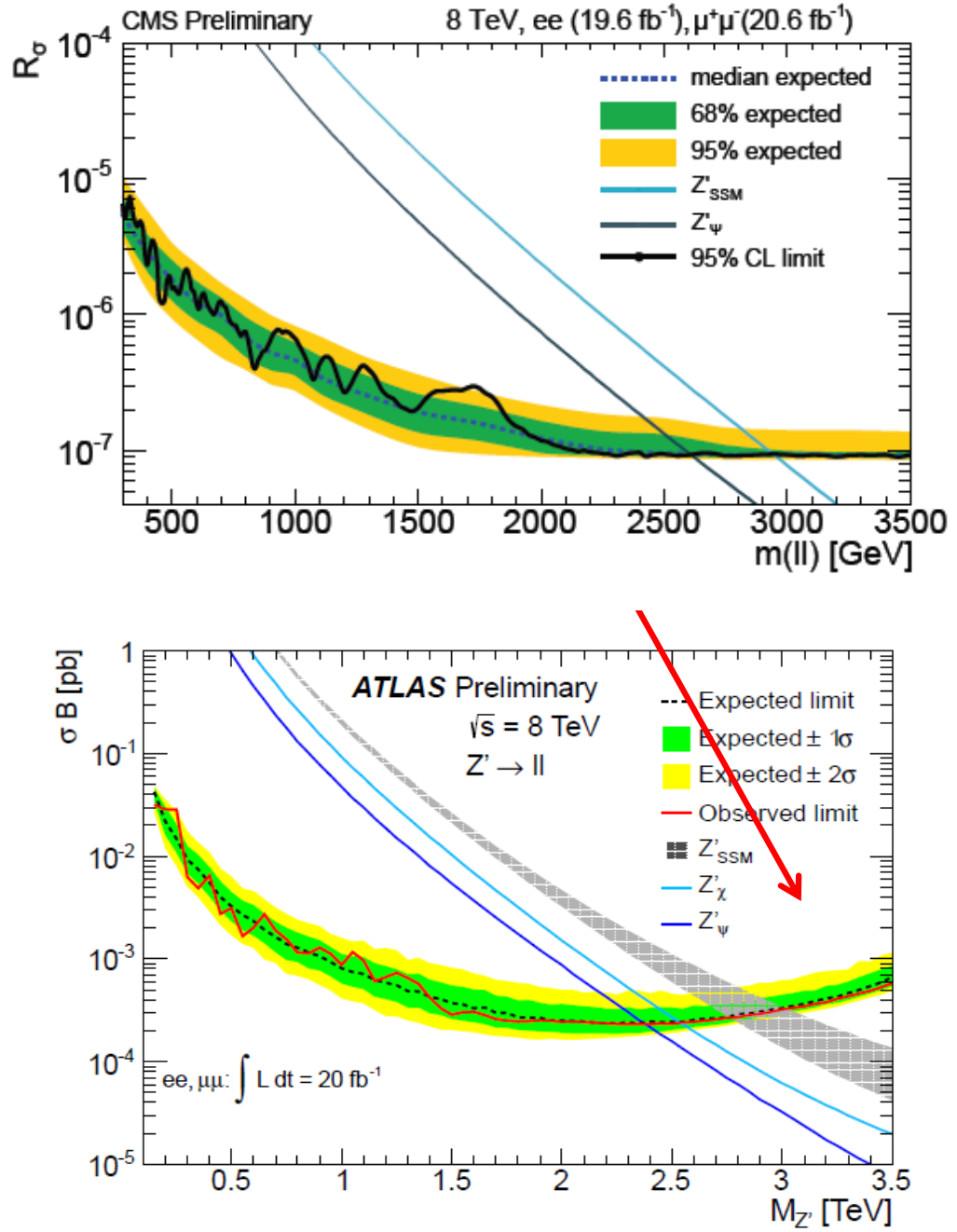


Figure 4.11: Comparison of ATLAS (bottom) and CMS (top) combined limit plots. The impact of the parton luminosity tail on limit curves is indicated by the red arrow. [50].

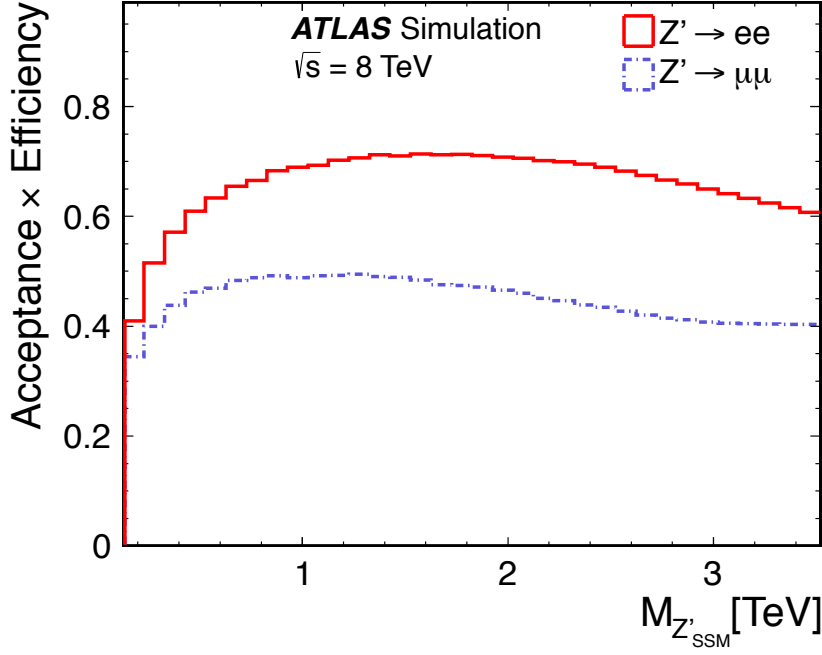


Figure 4.12: Product of acceptance and efficiency for the dielectron (red line) and dimuon (blue line) selections as a function of the Z'_{SSM} pole mass [20].

N_{tot} when solving for the new acceptance times efficiency, i.e.

$$(A\epsilon) = \frac{N_{\text{Sig}}}{N_{\text{tot}}} \quad , \quad (A\epsilon)' = \frac{N'_{\text{Sig}}}{N'_{\text{tot}}} \quad \Leftrightarrow \quad (A\epsilon)' = \frac{N'_{\text{Sig}}}{N_{\text{Sig}}} \cdot (A\epsilon) \quad (4.11)$$

where the primed notation denotes the remaining number of signal events in the modified templates and the correspondingly modified acceptance times efficiency fit. Since there is only a finite number of bins in the ROOT histograms, i.e. a finite number of ratios $N'_{\text{Sig}}/N_{\text{Sig}}$ that can be inferred from the templates, a polynomial fit of order six (eight) is used to obtain the new acceptance fit for the modified dimuon (dielectron) templates. The results are displayed in Figure 4.13 for the dimuon and Figure 4.14 for the dielectron channel. The diamond-shaped data points labelled “computed values” are the results of $N'_{\text{Sig}}/N_{\text{Sig}} \cdot (A\epsilon)$ used for the polynomial fits. As expected, the acceptance times efficiency of the modified templates (red line) is lower with respect to values of the original templates (blue line), especially in the high mass regime, since this is where parton luminosity tails become apparent (cf. Figure 4.7 and Figure 4.9). Accordingly, Z' exclusion limits computed using the modified templates will be less stringent as opposed to limit calculations using the original templates.

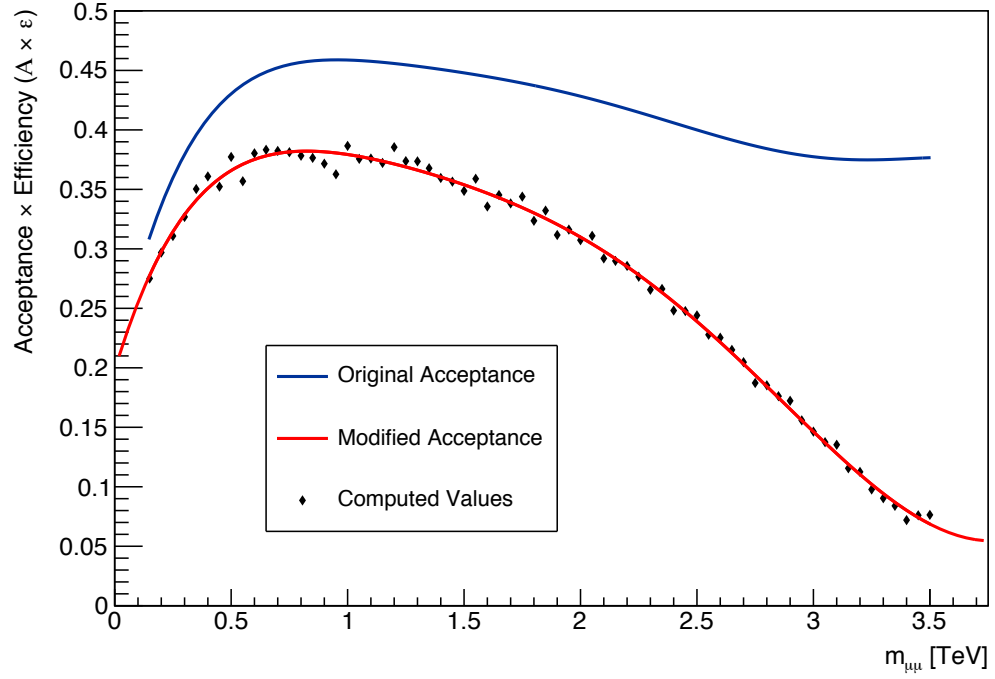


Figure 4.13: Original and modified acceptance $\times$ efficiency as a function of the dimuon invariant mass $m_{\mu\mu}$ at $\sqrt{s} = 8$ TeV.

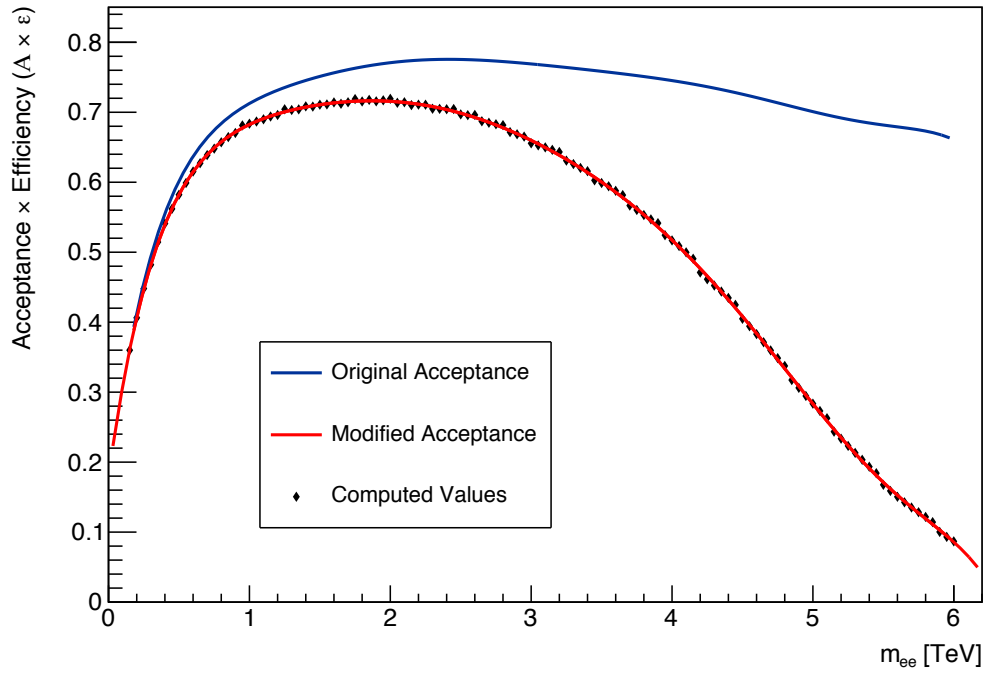


Figure 4.14: Original and modified acceptance $\times$ efficiency as a function of the dielectron invariant mass m_{ee} at $\sqrt{s} = 13$ TeV.

4.6 Limit Setting

Since no significant excess of events is observed, limits are placed on the cross section σ for Z' boson production times its branching ratio fraction B in individual dilepton channels and ultimately on the resonance mass $M_{Z'}$. As outlined in Section 4.5.1, each choice of Z' boson type and mass hypothesis corresponds to one specific signal template. Using Poisson statistics, a likelihood function to observe N_{obs} events can be defined (see [49]). By applying a marginalisation technique the dependence of the likelihood function can be reduced to one parameter of interest; in this case, it depends solely on σB . The marginalised likelihood function is [49]

$$\mathcal{L}'(N|\sigma B) = \int \mathcal{L}(N|\sigma B, \Theta_1, \dots, \Theta_N) d\Theta_1 \dots d\Theta_N \quad (4.12)$$

where $\Theta_1, \dots, \Theta_N$ are nuisance parameters that include uncertainties in any of the free parameters of the likelihood function. This function can be used to obtain the posterior probability density $\mathcal{L}'(\sigma B|N)$ by applying Bayes' Theorem [8]

$$\mathcal{L}'(\sigma B|N) = \mathcal{L}'(N|\sigma B) \frac{\pi(\sigma B)}{\pi(N)} \quad (4.13)$$

where $\pi(N)$ is the prior for the observed number of events and $\pi(\sigma B)$ is the prior for σB . The most likely signal content given the data is then given by the maximum of $\mathcal{L}'(\sigma B|N)$. The 95% CL Bayesian upper limit $(\sigma B)_{95}$ is obtained by integrating the posterior probability density, such that [49]

$$0.95 = \frac{\int_0^{(\sigma B)_{95}} \mathcal{L}'(\sigma B|N) d(\sigma B)}{\int_0^{\infty} \mathcal{L}'(\sigma B|N) d(\sigma B)} \quad (4.14)$$

A detailed description of the statistical methods employed in the analysis is beyond the scope of this thesis; a comprehensive overview is provided in [49].

After calculating 95% upper limits on σB , limit plots are generated using a ROOT macro that reads in all relevant data, including signal template and background files, as well as theoretical cross sections and k-factors. Lower limits on the Z' mass can be obtained from the upper limits on σB by finding their intersection with the theory curves as a function of $M_{Z'}$ for the different hypotheses under consideration [8]. The results for the dimuon channel at a centre-of-mass energy of $\sqrt{s} = 8$ TeV are summarised in Table 4.2, and the associated plots are shown in Figure 4.15. Note that the theory curves of Z'_ψ and Z'_χ belong to the smallest and largest σB values, respectively. Consequently, all other E_6 motivated theory curves would fall between those of Z'_ψ and Z'_χ . To prevent the cluttering of theory curves, they are not included in the plots.

Templates	Mass exclusion limits at 95% CL [TeV]		
	Z'_{SSM}	Z'_ψ	Z'_χ
	obs. (exp.)	obs. (exp.)	obs. (exp.)
original	2.516 (2.519)	2.042 (2.085)	2.225 (2.244)
modified	2.465 (2.493)	2.004 (2.057)	2.180 (2.224)

Table 4.2: Observed and expected lower limits on the mass of the benchmark Z'_{SSM} and the E_6 motivated Z'_ψ and Z'_χ at a 95% CL.

As expected (for reasons stated in Section 4.5.2), the limits placed using the modified templates are more conservative as compared to the limits placed using the original templates. It is furthermore noticeable that the limit curves in the plot corresponding to the calculations using modified templates (bottom) do not exhibit a flat tail at high masses comparable to that of plots prepared by the CMS collaboration. A CMS plot for the dimuon channel at $\sqrt{s} = 8$ TeV is shown for comparison in Figure 4.16. Interestingly enough, if the original acceptance fits are used instead of the modified acceptance fits, the shape of signal curves as it is to be seen in the CMS plots can be reproduced to an astonishing degree. If this is in fact the underlying cause for the altered shape of the curves with respect to Figure 4.17, it would imply that the modification of the signal templates is not accurate enough to reach a satisfactory degree of similarity between the modified templates used in this analysis and the true signal templates used by CMS, or that the acceptance fits used by CMS are not compatible with their signal shape assumptions. It goes without saying that there are various other possibilities to explain this behaviour, and even though the similarities between Figure 4.16 and Figure 4.17 are striking, these are only guesses that cannot be backed up on the basis of the available data.

The limit plots for the dielectron channel at $\sqrt{s} = 13$ TeV are shown in Figure 4.18. The ATLAS conventions for plot labels mentioned above apply here as well. Observed limit curves are not included in the dielectron plots. This, of course, reflects the fact that data taking at $\sqrt{s} = 13$ TeV is yet to commence. It should be noted that the theory curves of Z'_ψ and Z'_χ are intersecting at high mass (as can be seen at the very bottom of the plots). This is not supposed to happen and is indicative of an error. At the time of writing, the root of the problem is unclear as investigations are still ongoing. Seeing as it is not known if and how severely other elements of the calculations are affected, exclusion limits for the dielectron channel at $\sqrt{s} = 13$ TeV are not quoted in this thesis.

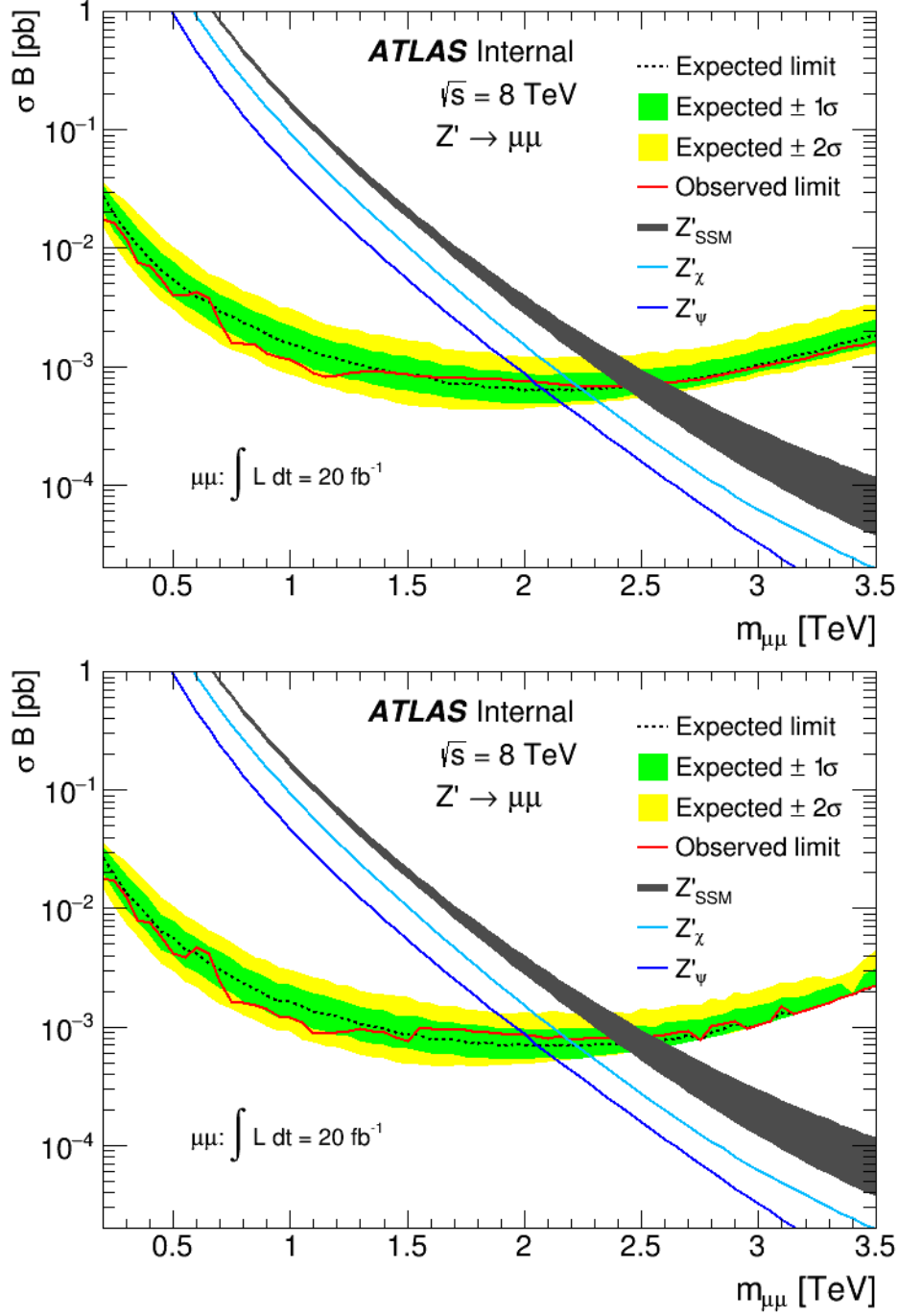


Figure 4.15: Observed (red line) and expected upper limits on σB for Z' boson production at $\sqrt{s} = 8 \text{ TeV}$ for the dimuon channel using original (top) and modified (bottom) signal template files. The median expected limit is plotted as a black dashed line, with $\pm 1\sigma$ (green) and $\pm 2\sigma$ (yellow) contour bands. The band of the Z'_{SSM} theory curve represents the theoretical uncertainty.

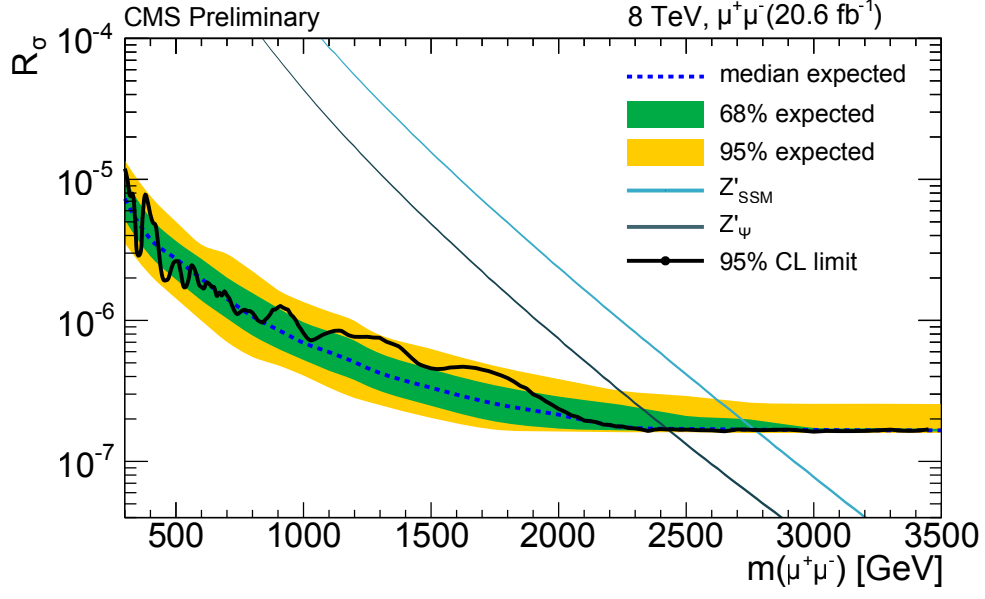


Figure 4.16: CMS upper limits on the production ratio R_σ of cross section times branching fraction for the dimuon final state as a function of resonance mass at $\sqrt{s} = 8$ TeV [51].

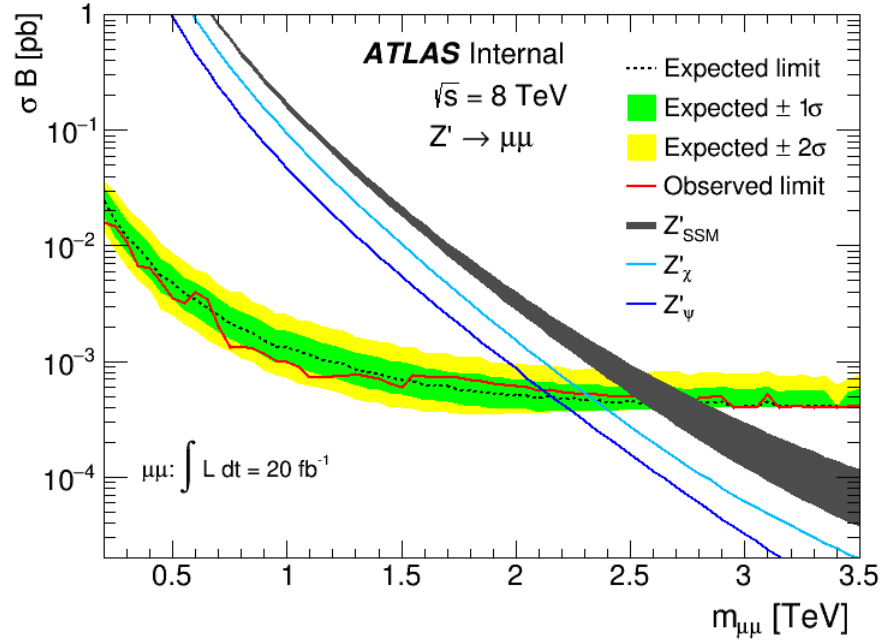


Figure 4.17: Upper limits on σB for Z' boson production at $\sqrt{s} = 8$ TeV for the dimuon channel using modified signal templates in combination with the original acceptance times efficiency fit.

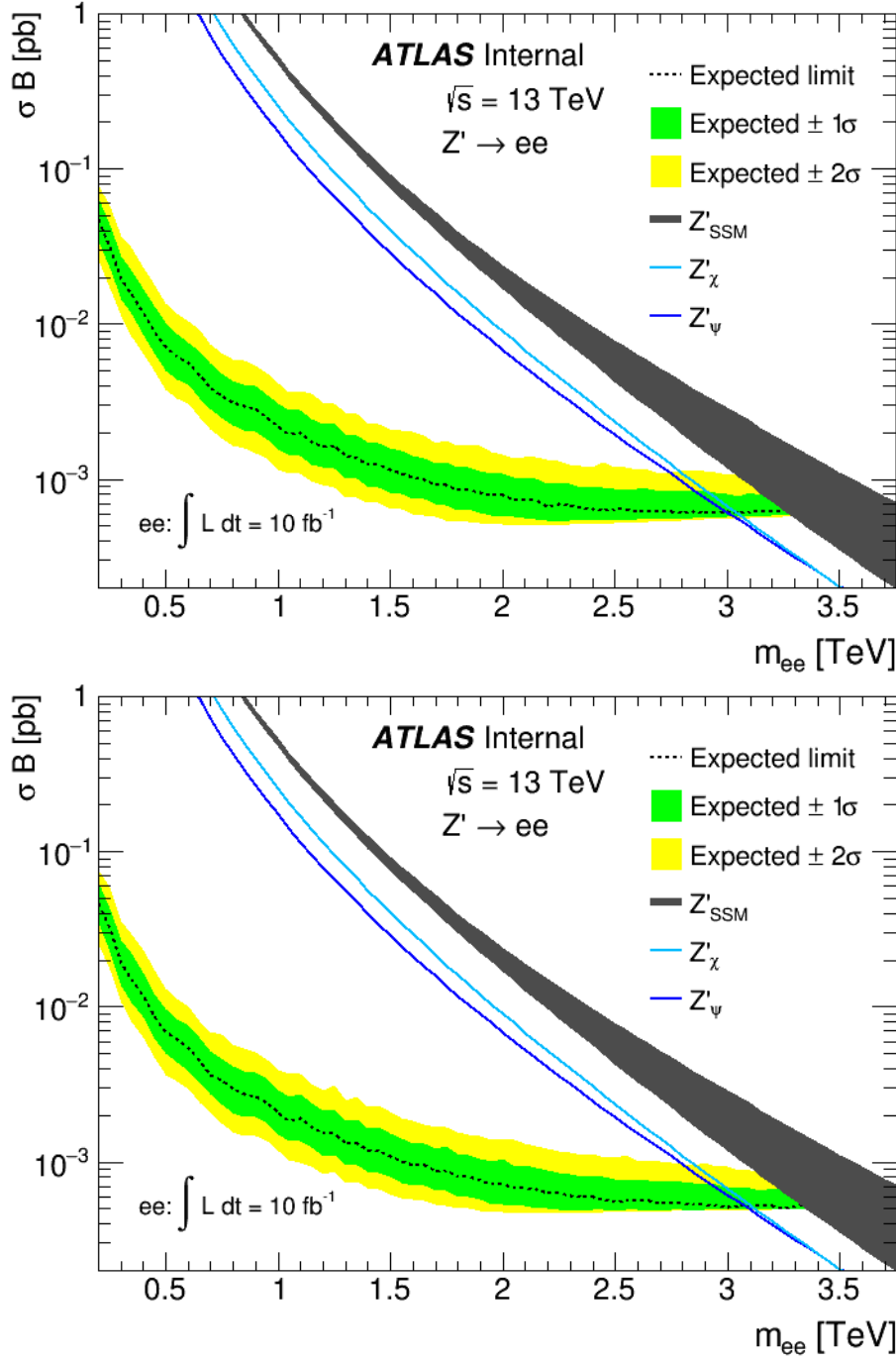


Figure 4.18: Expected upper limits on σB for Z' boson production at $\sqrt{s} = 13$ TeV for the dielectron channel using original (top) and modified (bottom) signal template files. The median expected limit is plotted as a black dashed line, with $\pm 1\sigma$ (green) and $\pm 2\sigma$ (yellow) contour bands. The band of the Z'_{SSM} theory curve represents the theoretical uncertainty.

Chapter 5

Conclusion

The ATLAS detector at the Large Hadron Collider is used to search for high-mass resonances decaying to dielectron or dimuon final states. A search for high-mass resonances decaying to dimuon final states was conducted using 20 fb^{-1} of pp -collision data collected in Run Ib of the LHC at $\sqrt{s} = 8 \text{ TeV}$ in the dimuon channel. The observed dimuon invariant mass spectrum $m_{\mu\mu}$ is consistent with the SM expectation. Upper limits are placed on the signal cross section times branching fraction σB for the benchmark Z'_{SSM} boson and E_6 motivated Z' bosons. The upper limits are then used to derive lower limits for the resonance mass $M_{Z'}$. A Z'_{SSM} is excluded below a resonance mass of 2.516 TeV at a 95% CL. The exclusion limits for the E_6 motivated bosons Z'_ψ and Z'_χ are 2.042 TeV and 2.225 TeV at a 95% CL, respectively. Following this procedure, expected limit plots were generated in the search for a high-mass resonance decaying to dielectron final states using data from simulated pp collisions at $\sqrt{s} = 13 \text{ TeV}$ in anticipation of Run II of the LHC.

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Appendices

Appendix A

ATLAS Coordinate Convention

It is common practice in particle physics to use a coordinate system tailored to the geometry of the detector, which - in the case of both ATLAS and CMS - exhibits cylindrical symmetry. The ATLAS collaboration uses a right-handed coordinate system, where the beam direction defines the z -axis. The centre of the detector is defined as the origin of the coordinate system and is referred to as interaction point. The positive x -axis is defined as pointing from the interaction point to the centre of the LHC ring and the positive y -axis as pointing upward. The x - y -plane is transverse to the beam axis z , hence it is often called transverse plane. Measurements of energy and momentum in the transverse plane are denoted by the subscript T , i.e. E_T and p_T , respectively. It is often more convenient to use polar coordinates (r, ϕ) to label points in the transverse plane, where $r = \sqrt{x^2 + y^2}$ is the radius and ϕ is the azimuthal angle around the z -axis defined from the positive x -axis. When using spherical coordinates (r, ϕ, θ) , the polar angle $\theta \in [0, \pi]$ is defined from the positive y -axis. Physical quantities measured in the transverse plane can be expressed in terms of θ . The transverse momentum p_T , for instance, is given by $p_T = p \sin \theta$.

Another useful coordinate is the pseudorapidity η , which is defined as

$$\eta := -\ln \left[\tan \left(\frac{\theta}{2} \right) \right]$$

It is a quantitative measure of a particle's angle relative to the beam axis z and is commonly used to specify the angular range that certain detector components are sensitive to. Using the definition of η , a Lorentz-invariant measure of angular separation can be defined:

$$\Delta R := \sqrt{(\Delta \eta)^2 + (\Delta \phi)^2}$$